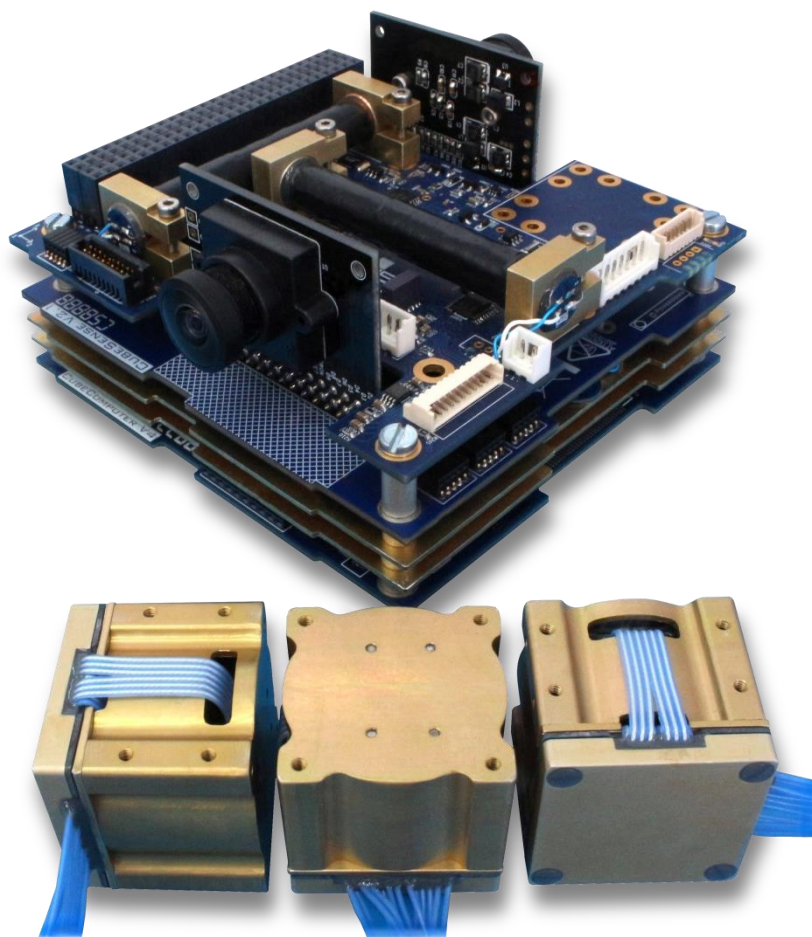




CUBEADCS

THE COMPLETE ADCS SOLUTION



INTERFACE CONTROL DOCUMENT

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List of Acronyms/Abbreviations

ABC	ADCS Body Coordinate
ACP	ADCS Control Program
ADCS	Attitude Determination and Control System
CSS	Coarse Sun Sensor
CPU	Central Processing Unit
EDAC	Error Detection and Correction
ESD	Electrostatic Discharge
I ² C	Inter-Integrated Circuit
MCU	Microcontroller Unit
MEMS	Microelectromechanical System
OBC	Onboard Computer
ORC	Orbit-Referenced Coordinate
PCB	Printed Circuit Board
RTC	Real-Time Clock
SBC	Satellite Body Coordinate
SEL	Single-Event Latch up
SEU	Single-Event Upset
TC	Telecommand
TLM	Telemetry
UART	Universal Asynchronous Receiver/Transmitter

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1 Introduction

CubeADCS provides attitude sensing and control capabilities to nanosatellites. The CubeADCS units consist of several CubeSpace modules integrated into a compact ADCS solution.

CubeADCS comes pre-programmed with firmware which allows the user to interface to the unit via one of multiple communication busses. The firmware allows the user to easily interface, update and extract information from the ADCS system. CubeADCS runs all the required ADCS estimators and controllers while monitoring the health of the sub-modules.

Ground support software is also supplied with each CubeADCS which allows the user to directly connect to the CubeADCS through a UART serial-to-USB cable using a PC.

This document describes the characteristics of the CubeADCS as well as the mechanical and electrical interfaces. CubeADCS can be configured with different sensors and actuators for different mission requirements. This document covers all CubeADCS configurations and readers can use their discretion in determining which parts are relevant to their various CubeADCS configurations.

If the reader has doubt over which parts are relevant, please mail us at ***info@cubespace.co.za***.

2 Handling

- **Anti-static**

CubeADCS contains a variety of static sensitive devices. Therefore, the appropriate electrostatic discharge (ESD) protection measures must be implemented. The unit must never be handled without proper grounding.

- **Cleanliness**

It is recommended that CubeADCS modules be handled in a clean environment. Therefore, an appropriate laminar flow workbench or a cleanroom of ISO class 8 or better, should be used.

- **Moisture**

The unit should be kept free of moisture or liquids, as these could have corrosive effects on the electronics and electronic joints, which may lead to degradation and loss of reliability of the circuits.

- **Shock**

The unit must be handled with care. Dropping or bumping a CubeADCS is prohibited.

- **Camera lens cleanliness**

The camera lenses should be kept clean and free of any dirt that may obstruct the images captured by the camera. Dust should be removed with a microfiber cloth. If required, the lens may be cleaned using ethanol and appropriate lens cleaning equipment. Unnecessary cleaning of the glass should be avoided.

- **Camera lens structural integrity**

The camera sensors are aligned to be parallel with the PCBs. This alignment is essential, as misalignment of the cameras influence the performance of the system. Therefore, external forces on the camera modules should be avoided.

- **Camera lens covers**

The optics are fitted with dust caps which should be removed before flight.

- **Reaction wheels**

The aluminum housings of the reaction wheels should NOT be tampered with. Tampering with the housings may damage the wheels. No attempt should be made to loosen or remove the fasteners that secure the housings.

3 Description and Overview

3.1 System components

CubeADCS can use a combination of magnetometer, coarse Sun sensor, fine Sun sensor, nadir sensor, MEMS rate sensor, and star tracker measurements to estimate the attitude of the satellite. Magnetorquers and reaction wheels are used to stabilize and control the satellite's attitude.

A CubeADCS unit consists of **up to four** integrated PC104-standard PCBs and several peripheral components, which are to be mounted separately. A basic diagram of the complete CubeADCS solution with all options is displayed in Figure 1. Older CubeADCS units are supplied with CubeSense as part of the Stack.

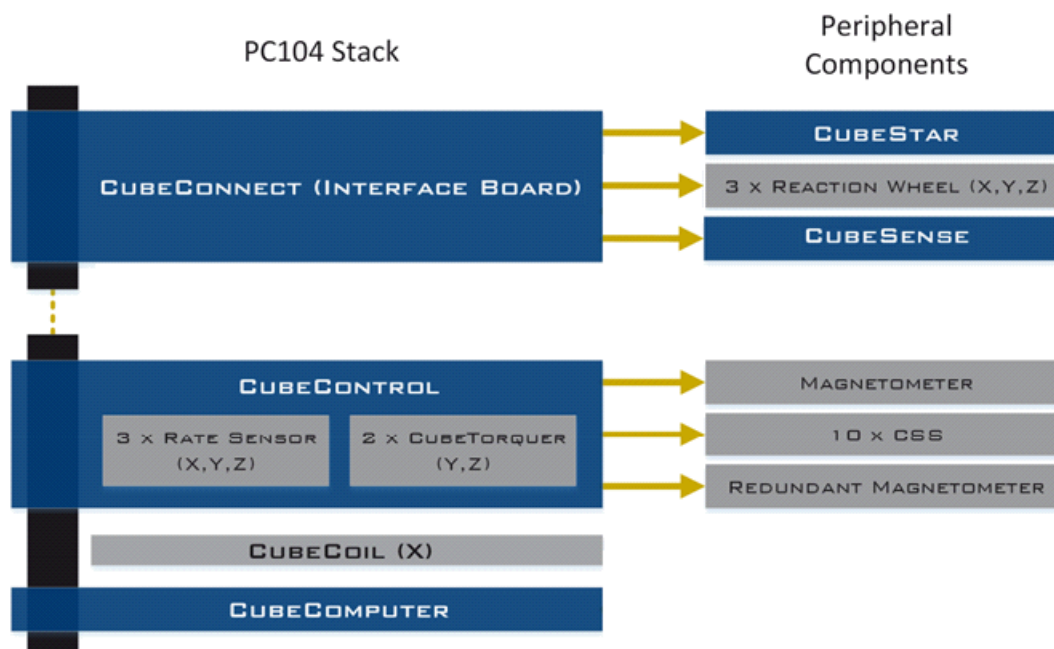
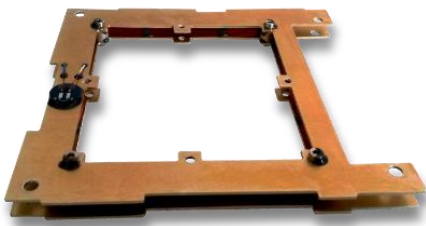
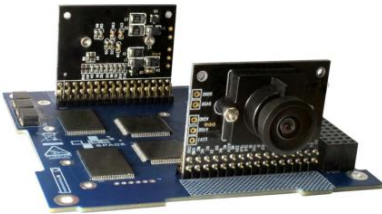


Figure 1 – System diagram of the complete CubeADCS solution.

A brief description of the core components of CubeADCS is given in Table 1.

Table 1 – Modules of the CubeADCS solution.

Module	Description
CubeComputer V4.2 	CubeComputer is a generic CubeSat OBC. Please see Section 7.1 for more information.
CubeCoil 	CubeCoil is a nanosatellite magnetic torquer in a low-profile air-coil form that slots into the stack between the PCBs. Please see Section 7.3 for more information
CubeSense V2 	CubeSense V2 is an integrated Sun and nadir sensor for attitude sensing Please see Section 7.2.1 for more information
CubeSense V3 	Compact version of CubeSense that consists of a single camera with full integrated electronics. Please see Section 7.2.2 for more information.

CubeControl V2.2



CubeControl is an actuator and sensor interface module. Please see Section 7.3 for more information.

CubeRod



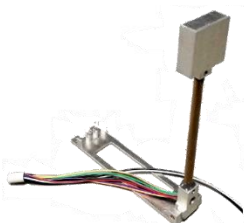
CubeRod is a nanosatellite magnetic torquer that uses a specially treated ferrous core with ultra-low remanence and high linearity. For more information please see Section 7.3.

CubeConnect

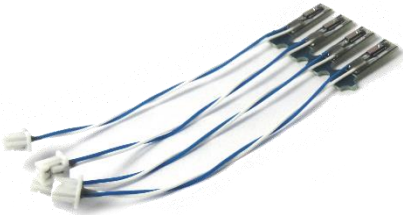


The CubeConnect board acts as an interface board for the wheels, CubeSense V3 modules and CubeStar module(s). For more information please see Section 7.6.

CubeMag



CubeMag is a 3-axis magnetometer. For more information please see Section 7.5.

Coarse Sun Sensors 	The coarse Sun sensors provide a coarse Sun vector throughout the entire sunlit part of the orbit. For more information see Section 7.6.
CubeWheel 	CubeWheel is a compact standalone reaction wheel unit for nanosatellites. Please see Section 7.8 for more information.
CubeStar V4.2 	CubeStar is a compact star tracker for nanosatellites. Please see Section 7.9 for more information.

3.2 Software

3.2.1 Bootloader

All CubeComputers delivered by CubeSpace have a bootloader pre-installed.

The bootloader performs several important tasks, such as

- Enabling internal and external watchdogs of the MCU preventing faulty or damaged applications from halting the CPU.
- Allowing reprogramming of application via UART or I2C interface
- Counting number of reboots and tracks boot attempts
- Providing status telemetry

3.2.2 Attitude Control Program (ACP)

The ADCS Control Program (ACP) is the application launched by the bootloader. It controls all functions on the CubeComputer and commands all the other CubeSpace subsystems within

the CubeADCS. It also coordinates the sampling and calibration of sensors and, execution of estimators and execution of control commands to the actuators.

3.2.3 ADCS Software Library

The ADCS software library implemented in the ACP has several standard estimation modes:

- MEMS Rate Filter
- Magnetic Rate Filter
- TRIAD
- Full-state EKF
- MEMS Gyro EKF

The ADCS software library also has several standard control modes:

- Detumbling (B-Dot)
- High rate detumbling
- Very high rate detumbling
- Y-Thompson
- Sun-spin
- Target tracking
- Sun-pointing

3.2.4 CubeSupport

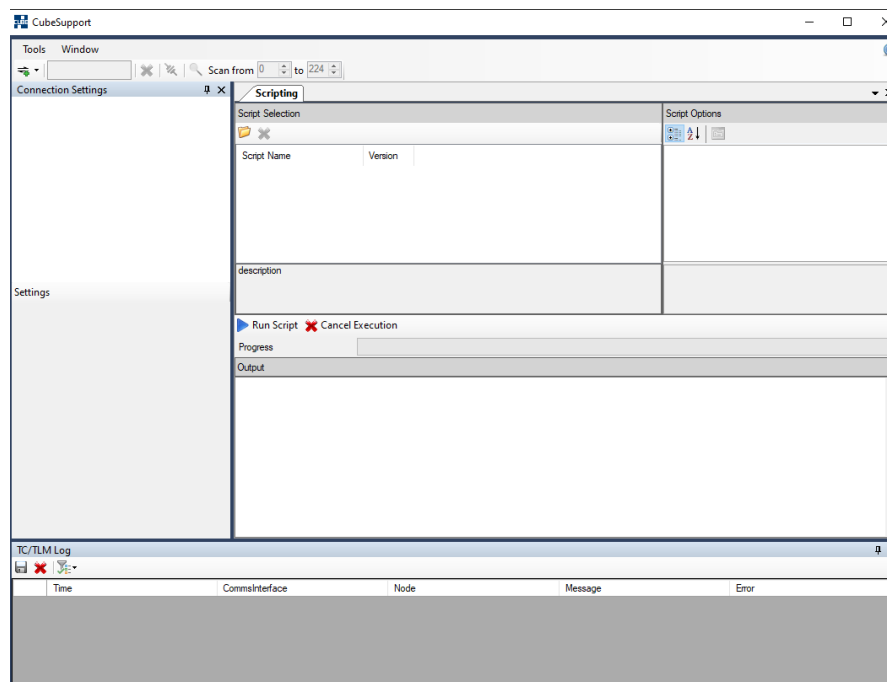


Figure 2 – CubeSupport

CubeADCS is also supplied with a ground support application called CubeSupport. This software runs on a PC and interfaces directly with the CubeADCS using a Serial-to-USB cable and a dedicated UART header found on the side of the CubeComputer.

The CubeSupport application can recognise and connect to the bootloader as well as the ACP. When connected to the bootloader, CubeSupport allows the user to upload new firmware to the external flash on CubeComputer for execution.

CubeSupport also provide the ability periodically log TLM, download images from imager-based sensors and to upload and alter the system configuration and ADCS configuration.

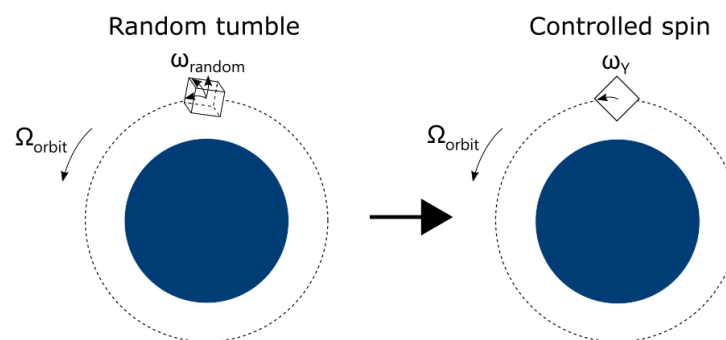
4 Standard CubeADCS solutions

4.1 Magnetic

The pure magnetic ADCS unit is typically used in 1U satellites but is not limited to them. In these satellites *accurate* pointing is rarely a requirement and space is very limited. This ADCS unit provides a robust, compact unit to get a 1U satellite from its initial random tumble, to a controlled slow spin.

Two CubeSense cameras can be added to the magnetic configuration to allow for more accurate attitude sensing and will allow the ADCS to detumble and perform slow spin-pointing manoeuvres.

The base version has small CubeTorquers mounted on CubeControl and is suitable for up to 3U sized satellites. For larger CubeSats, torquers can be upgraded to Medium or Large sizes for increased control torque.



4.1.1 Drawing

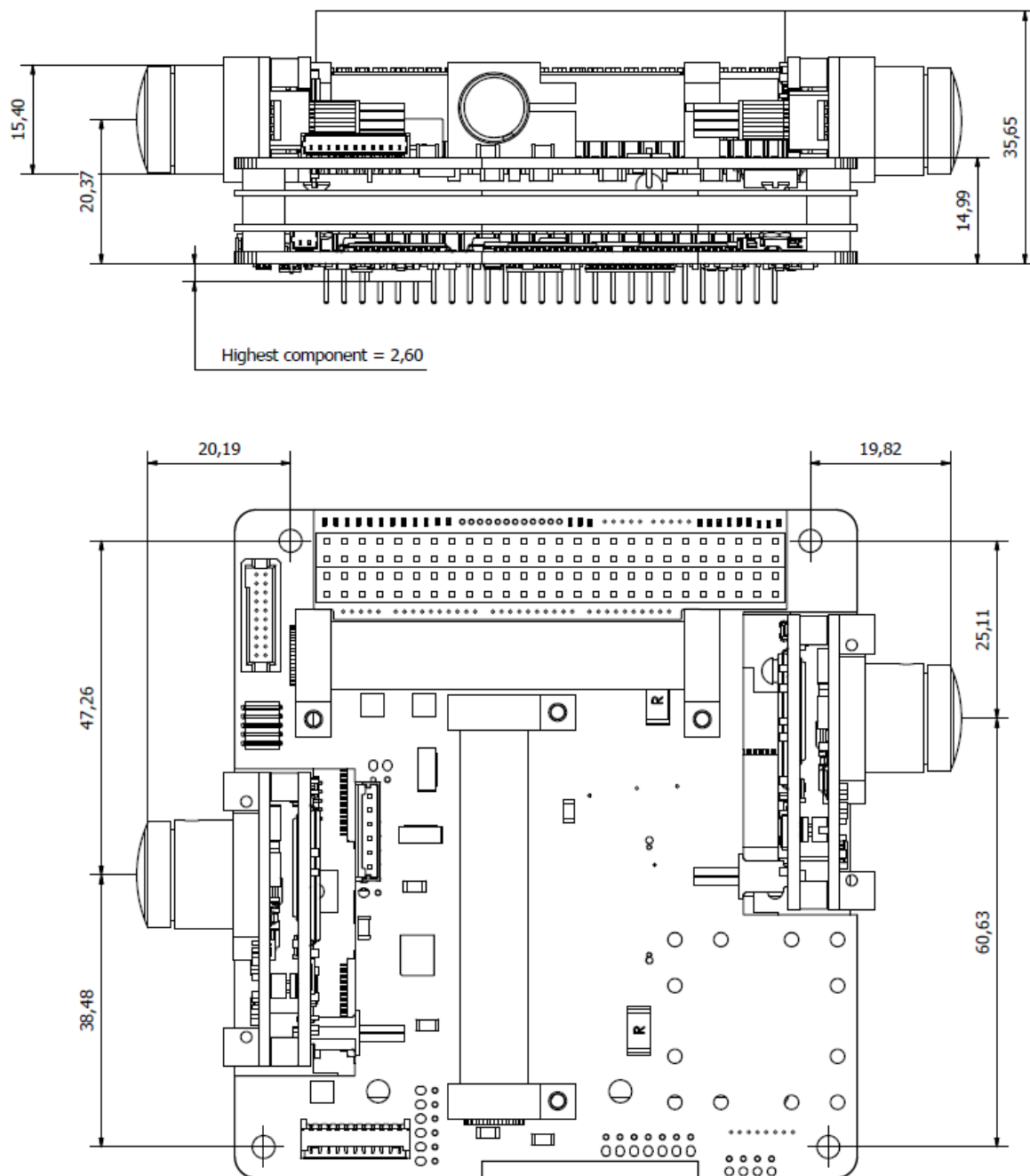


Figure 3 CubeADCS Magnetic with CubeSense V3

4.1.2 Specifications

Table 2 – CubeADCS Magnetic Specifications.

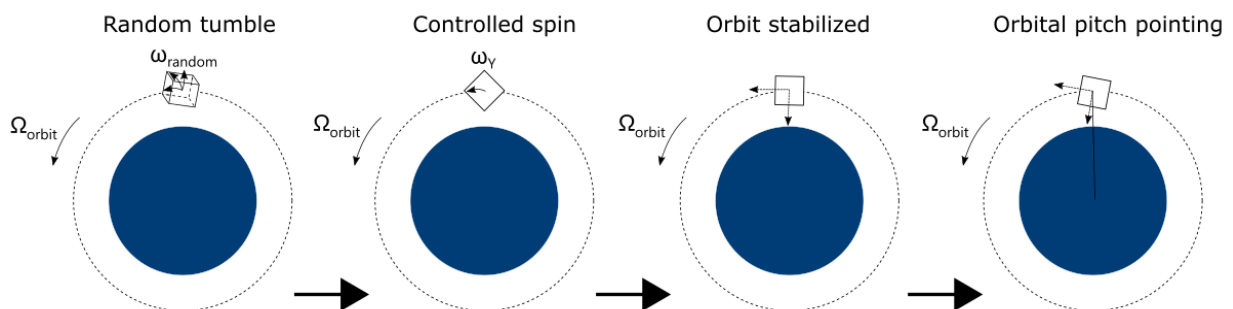
Specification	Value	Notes
Physical		
Mass	225 g	Without CubeSense
	300 g	With 2x CubeSense V3
Stack Dimensions	88.1 x 94.54 x 35.65 mm (0.36 U)	Without CubeSense
Power		
Power Supply	3.3 V, 5V, Vbat(6.5-16V)	
Power usage	445 mW	Average
	645 mW	Average (with CubeSense)
	1585 mW	Peak
	1985 mW	Peak with CubeSense
Performance		
Rate measurement accuracy	0.02°/s (3σ)	
Attitude measurement accuracy	5°(3σ)	Without CubeSense
	0.6°(3σ)	With CubeSense in sunlight part of orbit
Environment		
Operating temperature range	-10°C to +60°C	
Vibration	8.03 g RMS random	
Radiation	20 kRad	
Components		
Included	CubeComputer	
	CubeControl with 10x Coarse Sun Sensors	
	2x CubeRod 1x CubeCoil	
	CubeMag	
Optional	Redundant Magnetometer	
	CubeSense V3	

4.2 Y-Momentum

For the majority of CubeSat missions, the payload ADCS requirement is to be 3-Axis stabilized, with the payload pointing nadir or in the flight direction.

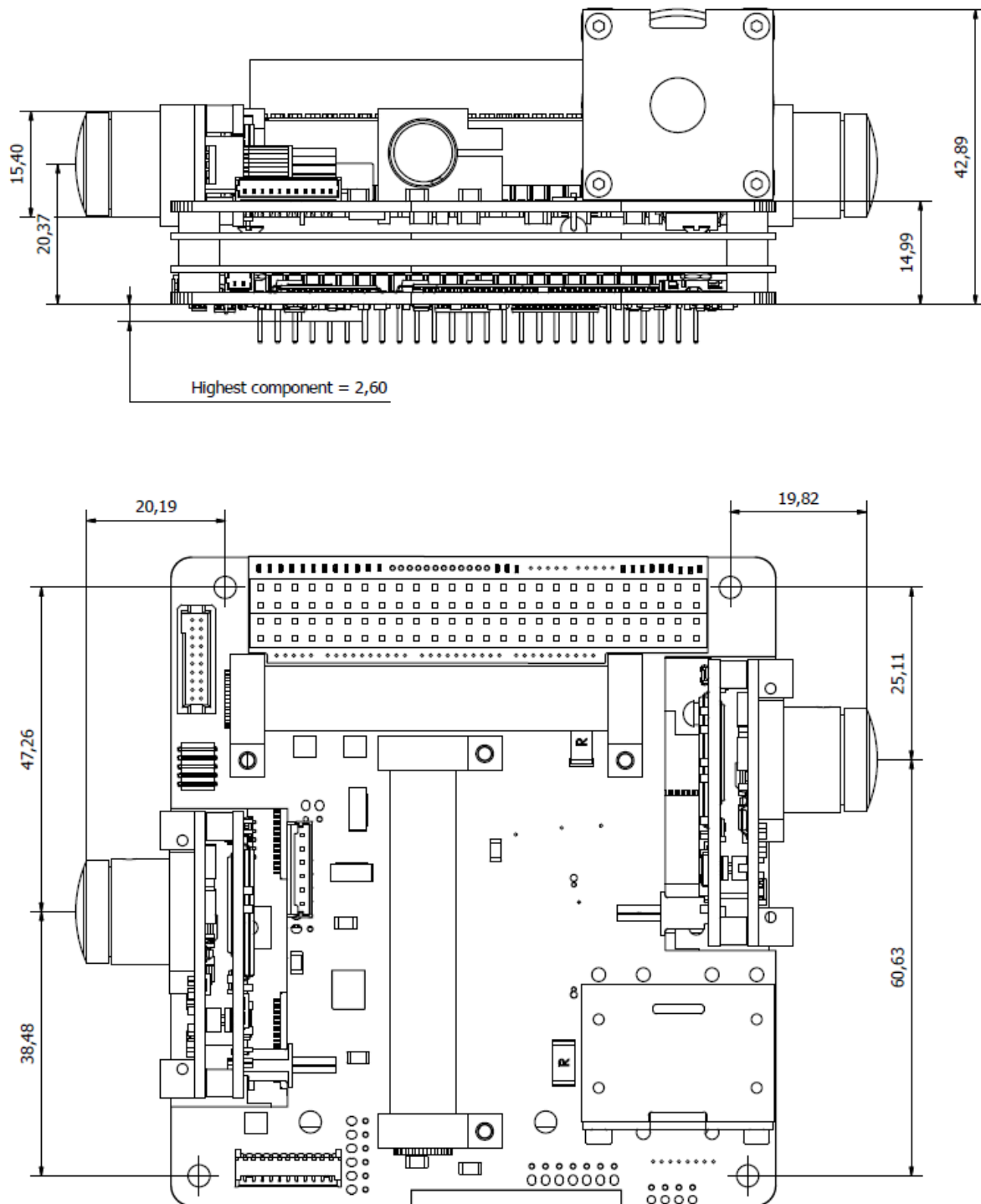
4.2.1 Typical application

The Y-Momentum bundle is used for missions where $<1^\circ$ pointing is not necessarily needed, but a 3-axis stable system is required. Examples of such missions are general Earth observation missions, scientific missions where the payload must be pointed in the velocity direction, and communication missions. This ADCS unit provides the functionality to detumble a satellite into a stable spin and then to absorb the rotation of the satellite into a momentum wheel. This momentum wheel provides good disturbance rejection which makes it possible to further keep the satellite 3-axis stable using magnetic control. The satellite can do pitch maneuvers in the orbital plane by controlling the momentum wheel's speed. This is particularly useful for imaging missions and ground station tracking in communication missions. For any other mission, ground station communications can be improved by doing tracking at least in the pitch direction.



The base version of the Y-Momentum system has a CubeWheel Small momentum wheel and small torquer rods and is suitable for up to a 3U. For larger satellites the torquers as well as the momentum wheel can be scaled up to suit the MoI of the satellite. In such a case, the larger actuators are not mounted on the PC104 stack anymore, but rather mounted separately on the satellite structure by the user.

4.2.2 Drawing



4.2.3 Specifications

Table 3 Y-Momentum CubeADCS Physical Specifications (Small versions)

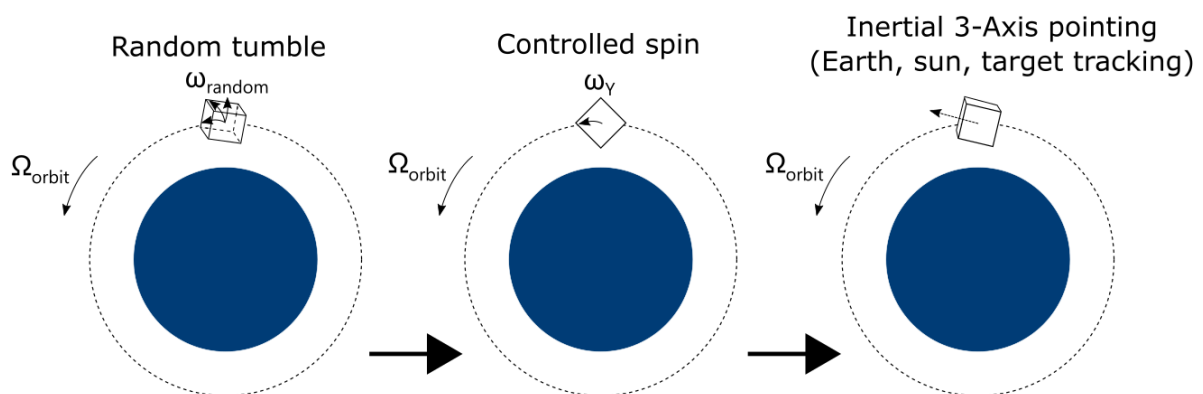
Specification	Value	Notes
Physical		
Mass	300 g	Without optional components
Stack Dimension	90 x 96 x 42.89 mm (0.43 U)	Without optional components
Power		
Power Supply	3.3V, 5V, VBat (6.5V -16V)	
Power consumption	571 mW	Average, Without optional components
	2295 mW	Peak, Without optional components
Performance		
Rate measurement accuracy	0.02°/s (3 σ)	
Attitude measurement accuracy	5°(3 σ)	Without CubeSense.
	0.6°(3 σ)	With CubeSense in sunlight part of orbit
Attitude control accuracy	10°(3 σ)	
	3°(3 σ)	With CubeSense in sunlight part of orbit
Environment		
Operating temperature range	-10°C to +60°C	
Vibration	8.03 g RMS random	
Radiation	20 kRad	
Components		
Included	CubeComputer	Running ACP
	CubeControl	3 x MEMS Gyro, 10 X Course Sun Sensors, 2x CubeRods and 1x CubeCoil
	CubeMag	
	CubeWheel Small	
Optional	Redundant Magnetometer	
	CubeSense	
	Upgrade to CubeWheel Medium or Large	
	Replace the CubeCoil with another CubeRod	
	Increase torquer windings	Typically for 6U missions

4.3 3-Axis

CubeSats have evolved to the point where high accuracy inertial pointing has become a requirement in certain missions. In these missions, three reaction wheels are required to provide 3-axis control throughout the entire orbit. Sensors are chosen to provide the level of accuracy required for a specific mission. This system can be upgraded to a variety of levels to provide different pointing abilities for different missions. With the latest stand-alone fine sun sensor and nadir sensor, the 3-Axis system is available in a highly compact low-profile unit.

4.3.1 Typical Application

The 3-Axis ADCS solution is designed to do inertial pointing with high accuracy. It contains all the controllers that are included in the other ADCS solutions but provides the extra functionality of 3-Axis pointing in any direction. The unit can function with or without the star tracker, depending on the pointing accuracy that is required. If high accuracy pointing is only required in the sunlit part of the orbit, a star-tracker is not required, and the system can function sufficiently with only the fine Sun and nadir sensor.



As with the Y-Momentum system, the base version contains small torquers and reaction wheels, which are mounted on the PC104 stack. For larger satellites requiring larger actuators, the reaction wheels and torquers are mounted on the satellite structure by the user. CubeSpace can assist with designing mounting brackets if required.

4.3.2 Drawing

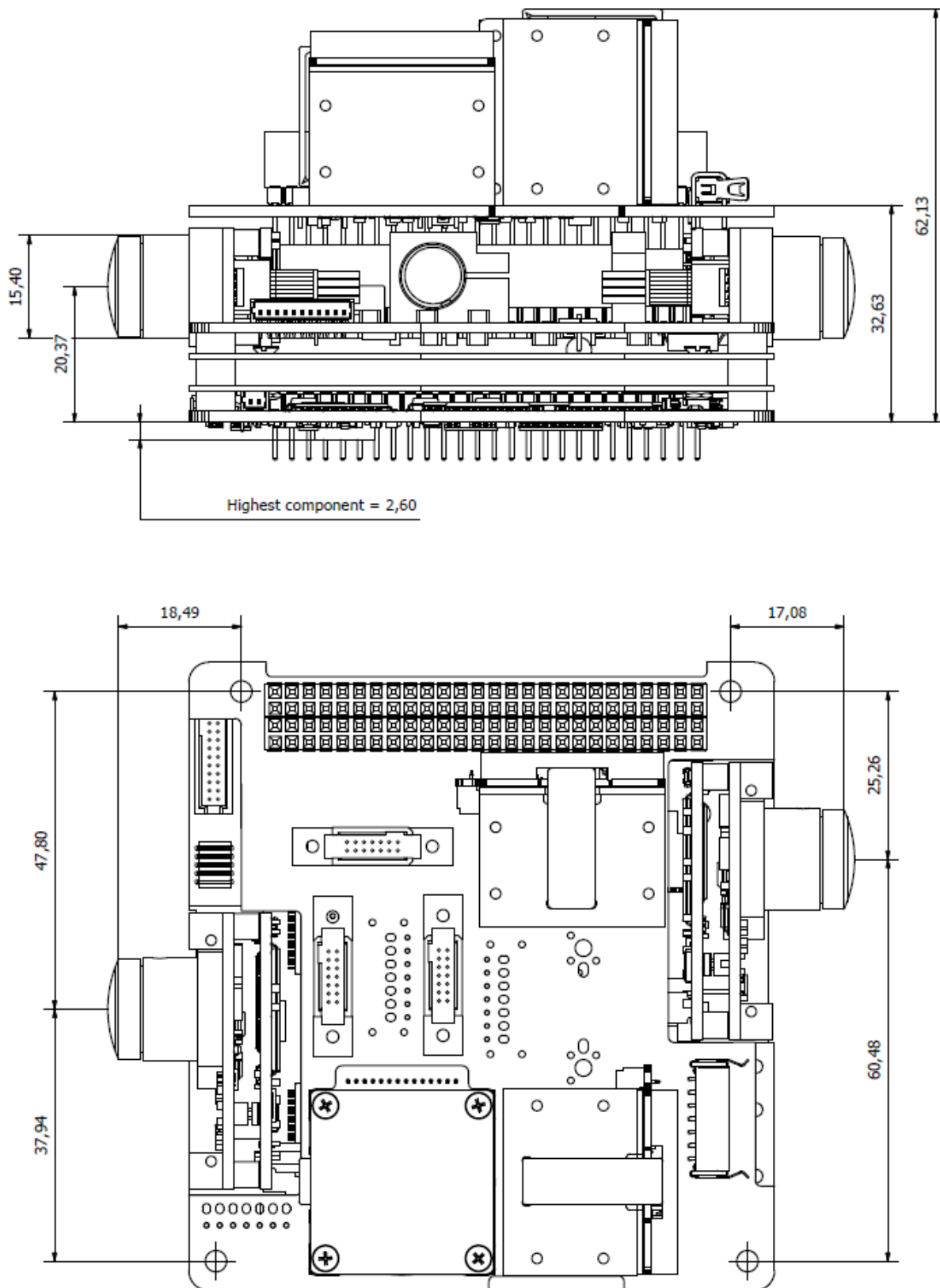


Figure 4 - Drawing of 3-Axis unit with mounted wheels (small version)

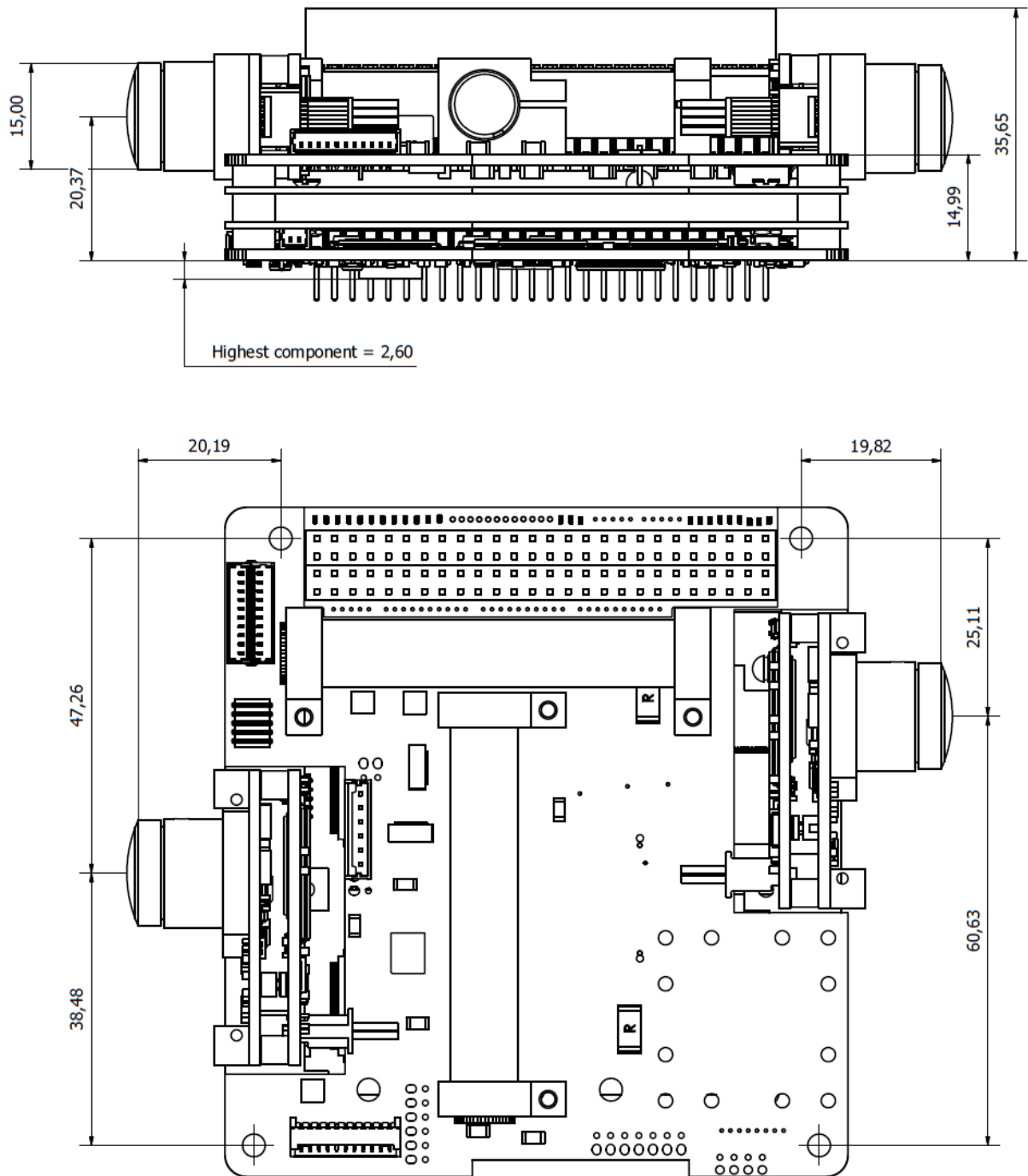


Figure 5 - Drawing of low-profile 3-Axis unit with wheels not mounted on stack

4.3.3 Specification

Table 4 – 3-Axis CubeADCS Specifications.

Specification	Value	Notes
Physical		
Mass	328 g	Without optional components
	554 g	With Small CubeWheels on CubeConnect
Stack Dimension	90 x 96 x 52 mm (0.52 U)	Without optional components
	90 x 96 x 75 mm (0.75 U)	Small CubeWheels mounted on CubeConnect, above CubeControl
Power		
Power Supply	3.3V, 5V, VBat (6.5V -16V)	
Power consumption	571 mW	Average, excl. wheels and optional components
	2295 mW	Peak, excl. wheels and optional components
Performance		
Rate measurement accuracy	0.02°/s (3 σ)	Assuming uniform symmetrical 2U CubeSat with no deployable
Attitude measurement accuracy	5°(3 σ)	Without CubeSense.
	0.6°(3 σ)	With CubeSense in sunlight part of orbit
	0.1(3 σ)	With CubeStar
Attitude control accuracy	6°(3 σ)	Assuming uniform symmetrical 2U CubeSat with no deployable
	1°(3 σ)	With CubeSense in sunlight part of orbit
	0.2°(3 σ)	With CubeStar
Environment		
Operating temperature range	-10°C to +60°C	
Vibration	8.03 g RMS random	
Radiation	20 k Rad	
Components		
Included	CubeComputer	
	CubeControl	3 x MEMS Gyro, 10 X Course Sun Sensors, 2x CubeRods and 1x CubeCoil
	CubeSense	
	CubeMag	
	3x CubeWheel Small	

Optional	Redundant Magnetometer	
	CubeStar	
	Upgrade to 3x CubeWheel Medium or Large	
	Replace the CubeCoil with another CubeRod	
	Increase torquer windings	Typically for 6U missions

5 Electrical Interface

5.1 PC104 interface

The CubeADCS mainly makes use of the standard PC104 header for electrical interfacing. The pin description of the PC104 bus is shown in Figure 6, as used by the CubeADCS. Where the row specified as "option" means that only some of the PC104 pins will be used for that row. These options are configured during manufacturing as specified in the Hardware Configuration document.

H2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51
H1	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51

PC104 interface pins					
Communication					
H1	1	CANL	CAN bus low		option
H1	3	CANH	CAN bus high		option
H1	21	I2C_SCL_ADCS	Internal I2C clock for all ADCS modules		required
H1	23	I2C_SDA_ADCS	Internal I2C data for all ADCS modules		required
H1	41	I2C_SDA_SYS	System I2C data for CubeComputer		required
H1	43	I2C_SCL_SYS	System I2C clock for CubeComputer		required
H1	17, 18, 19, 20	UART_1	Usable pins for UART_1 (RX or TX)		option
H2	21, 22	UART_1	Usable pins for UART_1 (RX or TX)		option
H1	33, 35, 39, 40	UART_2	Usable pins for UART_2 (RX or TX)		option
H1	29	SPI_CLK	SPI Clock		option
H1	30, 31	SPI_MOSI / MISO	SPI MOSI or MISO		option
H1	32	SPI_CS	SPI Chip Select		option
Power					
H2	29, 30, 32	GND	Ground connection for all modules		required
H2	45, 46	V_Bat	Battery voltage bus		required
H2	25, 26	5V_Main	Main 5 V supply		standard option
H2	27, 28	3V3_Main	Main 3.3 V supply		standard option
H1	47, 49, 51	5V_S	Switched 5 V supply options		option
H1	48, 50, 52	3V3_S	Switched 3.3 V supply options		option
H2	42	BUVIN	CubeComputer optional backup power supply		option
Internal ADCS pins					
H1	2, 4, 6, 8, 11	ENABLE	Enable lines for CubeADCS modules position 1		standard option
H1	16	ENABLE	Enable lines for CubeADCS modules position 2		option
H2	17, 18, 19, 20				
H1	5, 7, 9	ENABLE	Enable lines for CubeWheel 1 - 3		option
H1	13, 14, 15	ENABLE	GPIO/ ADC or Enable Lines		option
H1	10	ENABLE	Enable Line for CubeStar position 1		option
H2	15	ENABLE	Enable Line for CubeStar position 2		option

Figure 6 – PC104 bus pin description.

5.2 Power

5.2.1 Power supply

The ADCS must be supplied with regulated 3.3 V, 5.0 V and the raw battery voltage, V_{battery} . Battery voltage must be between 6.5V and 16V.

The 3.3 V and 5.0 V buses must be provided by one of H1-48, 50, 52 or H2-27, 28 (3.3 V) and one of H1-47, 49, 51 or H2-25, 26 (5 V) – **depending on the selection made by user in the Hardware Configuration Sheet**. The 3.3 V and the 5 V supply must be switched on within 10ms of each other.

The battery bus (pins H2-45 and H2-46) supplies power to the CubeWheels and the magnetometer boom deployment circuitry.

5.2.2 Inrush Currents

The CubeADCS and its sub-components have been optimised to limit the effect of inrush currents. However, the power supply, and the connection from the power supply to the CubeADCS, can still introduce inrush problems if not set-up correctly. Therefore, it is important to ensure:

- Power supply can output at least 12W (1.5A at 8V). During ground testing, 1A at 8V will suffice, as the magnetometer will not be deployed.
- Cables between 3.3 V power supply and CubeADCS are short with sufficient thickness. Ideally <15cm and <24 AWG.

If these conditions are met, inrush should not pose a problem during ground testing or orbit.

5.3 Enable lines

The ADCS has internal power switches for the various individual components (CubeSense, CubeControl, CubeStar and the CubeWheels), all of which are controlled by CubeComputer. **Figure 7** shows an example of the power switching connections. The switch states are available as telemetry (TLM) and can also be toggled using a telecommand (TC).



Please note that the enable lines are connected straight to the CubeComputer's MCU. This means that these pins on the MCU are directly exposed to the PC104 header. If any overvoltage signal inputs are applied to the MCU it will damage and possibly destroy the MCU.

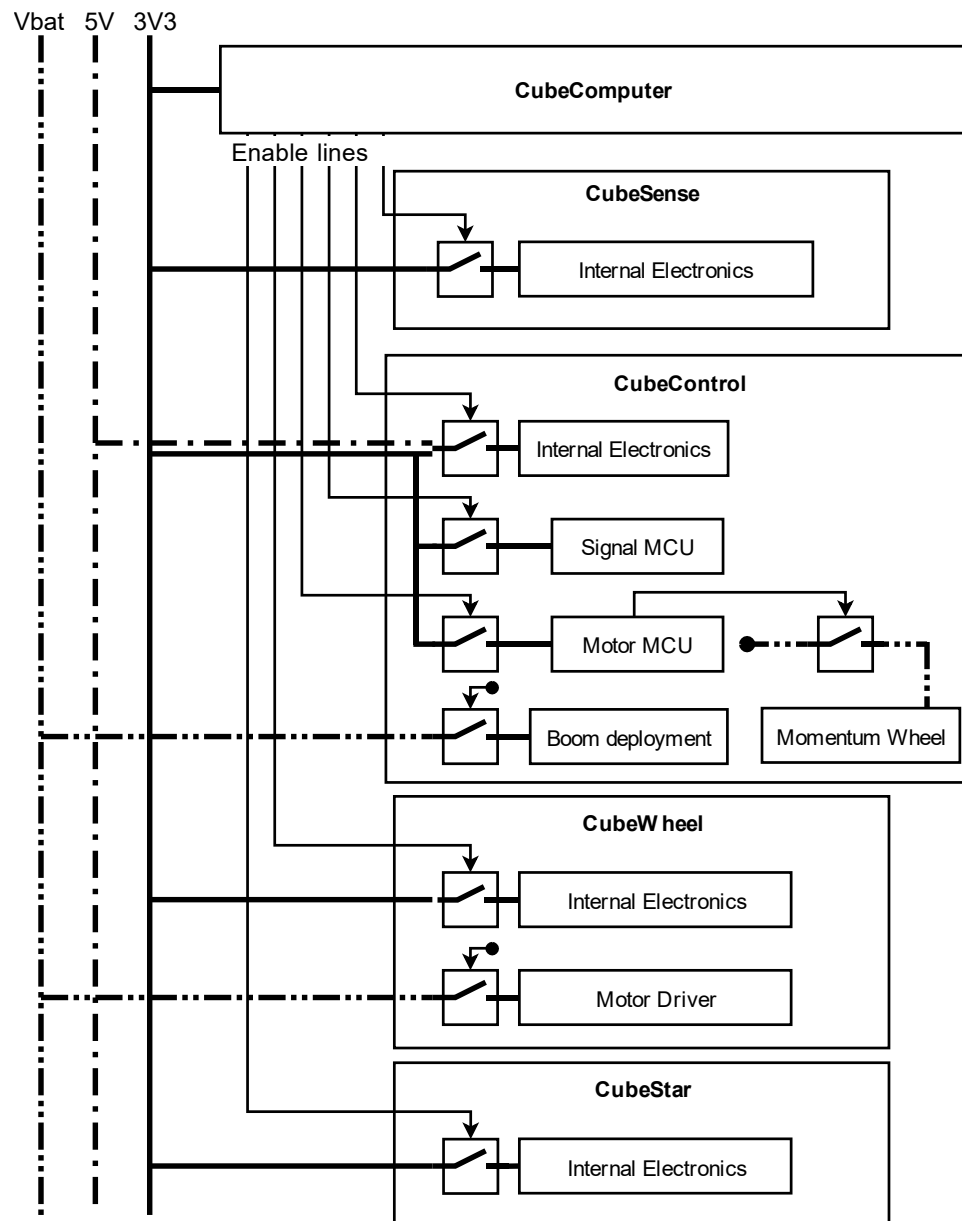


Figure 7 - CubeADCS Enable Lines

5.4 Communication

5.4.1 I²C

The CubeADCS unit can communicate with other satellite subsystems using the system I²C on the PC104 bus (H1-41, 43). CubeComputer will act as a slave on the system I²C bus, responding to commands and telemetry requests. CubeComputer can be populated with or without pull-up resistors on the system I²C bus. The system I²C is permanently connected to H1-41,43,

making this the primary communication channel for CubeComputer and the CubeADCS. All I²C lines on the CubeComputer are buffered using the [PCA9512ADP,118](#) from NXP.

5.4.2 UART

The CubeADCS bundle has two UART channels (on CubeComputer), designated UART 1 and UART 2, which can be used to interface with the CubeADCS. Both channels can be connected to the PC104 header. Both UART lines on the CubeComputer are buffered using the [SN74AVC2T45](#) from Texas Instruments.

CubeSupport communicates with CubeComputer via UART 1, which is also accessible via a 3-way 2.54 mm pitch square post female header. A UART-to-USB cable is supplied with each CubeADCS unit, which enables the user to connect with a PC to the CubeADCS.

5.4.3 CAN

The optional CAN interface on CubeComputer is only available on the main PC104 header (H1-1, 3). The combination of a CAN transceiver and a CAN controller module on CubeComputer allows the CubeADCS to interface at CAN bus voltage levels of 3.3 V or 5 V. A termination resistor between the CANH and CANL lines can be populated and can be configured at order.

6 Mechanical Interface

6.1 Component stack and shape

Most CubeSpace components make use of the standard “self-stacking” PC104 CubeSat mechanical interface. The CubeADCS stack includes spacers to support the inter-PCB spacing. The PC104 header at the top of the stack is the ESQ-126-38-G-D, and the length of the pins at the bottom of the stack is 5.75 mm. Please see Section 4 for examples of our standard CubeADCS configurations and their respective heights.

The dimensions of the stacked components and mounting locations are displayed in Figure 8 and Figure 9. It should be noted that there are two different variants. The new CubeADCS products will have rounded corners (from start of 2020 and onwards). The legacy square shape is included for sake of completeness. If there is uncertainty about which shape is applicable for a specific product, please contact the CubeSpace team.

The height of typical ADCS solutions can be found in Section 4.

6.1.1 Rounded PCB Shape

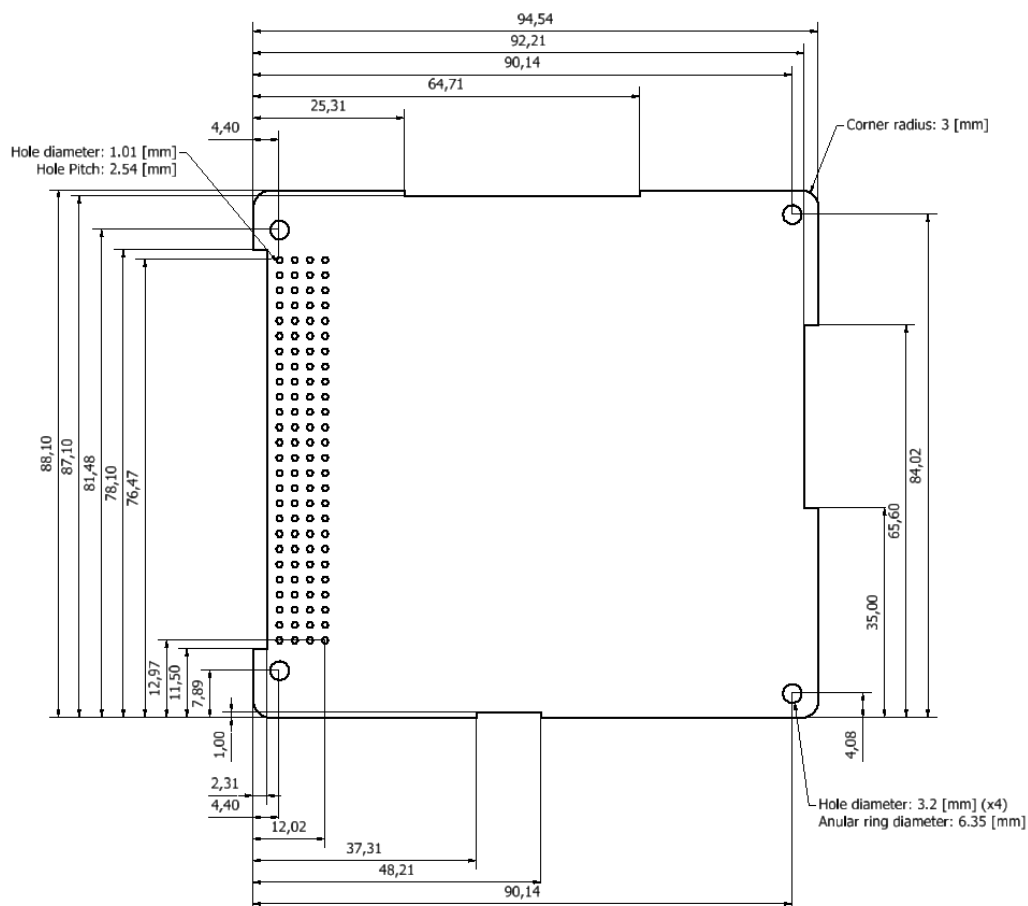


Figure 8 – PC104 board with rounded corners.

6.1.2 Square PCB Shape

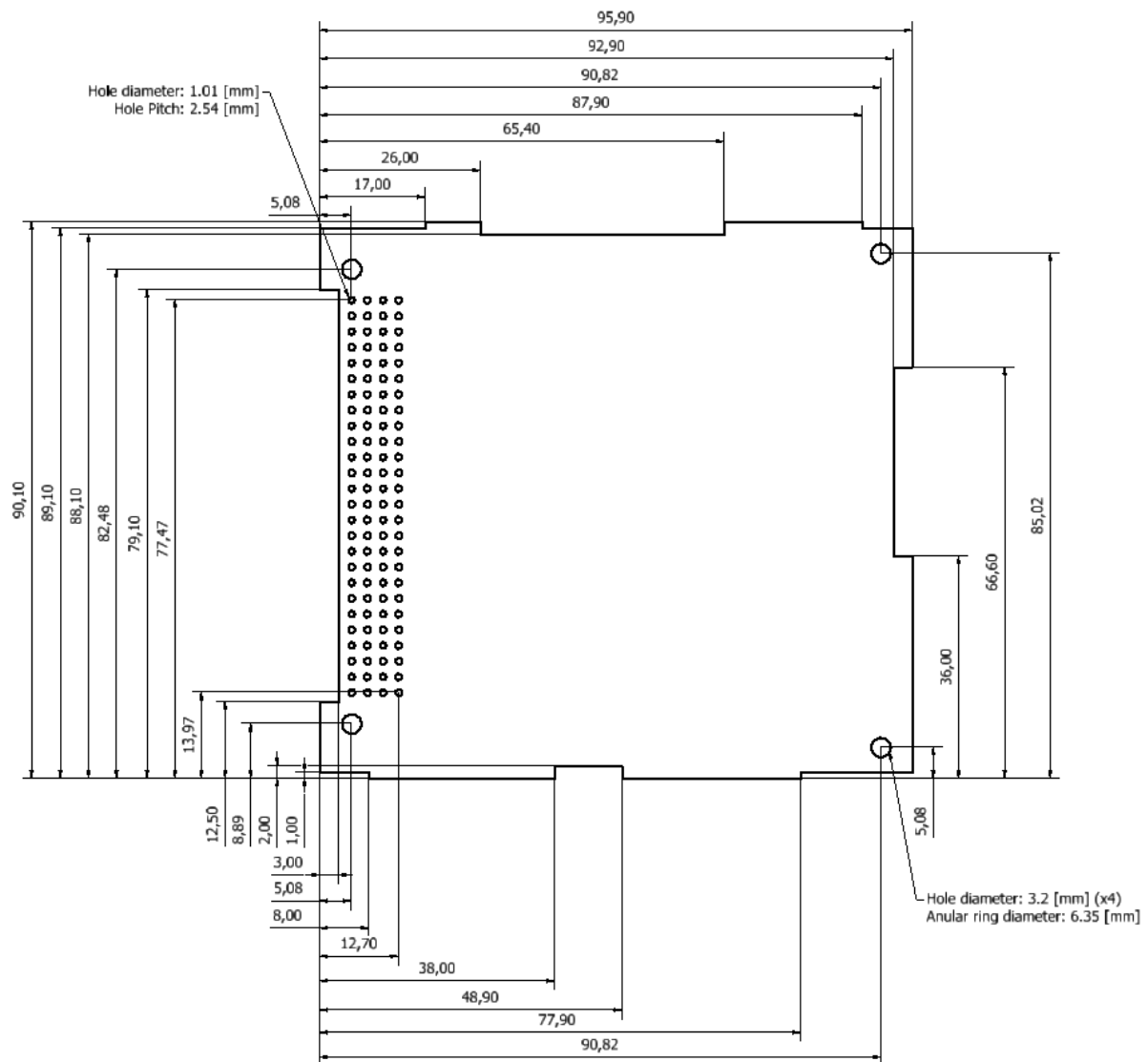


Figure 9 – Standard PC104 board.

The individual components (PCBs) of the ADCS are connected to one another through the main PC104 header. For additional structural support the PCBs are supported by aluminium spacers and kept in place by M3, A4 stainless steel bolts and nuts through the mounting holes. When the ADCS is integrated with the rest of the satellite these nuts and bolts will have to be removed. This will cause the spacers to easily fall out of the ADCS. To prevent this, the spacers are epoxied in place to insure they remain in position. Only the bottom part of each spacer will receive epoxy, as shown in Figure 10.

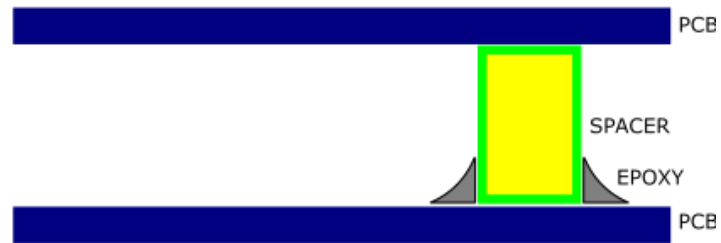


Figure 10 – Epoxy of spacers

7 CubeComponents

7.1 CubeComputer

The controlling unit in the ADCS stack is the CubeComputer a low power OBC. CubeComputer uses a high-performance, low-power 32-bit ARM Cortex-M3 based MCU. The computer was developed for ADCS processing. It runs estimator and control algorithms, does logging of TLM and manages communication to the ADCS modules. The CubeComputer features the following functions as standard:

- An integrated RTC and an internal and external watchdog
- A 32 kB EEPROM is used for the firmware (bootloader)
- A 4 MB Flash for the code and in-flight reprogramming
- A FPGA for flow-through EDAC and SEU protection
- Current monitoring for SEL (latch-up) protection and power cycling ability
- A MicroSD card for 1 GB of data storage
- I2C, CAN, UART interfaces to another OBC

The UART header described in Section 5.4.2 is shown in Figure 11.

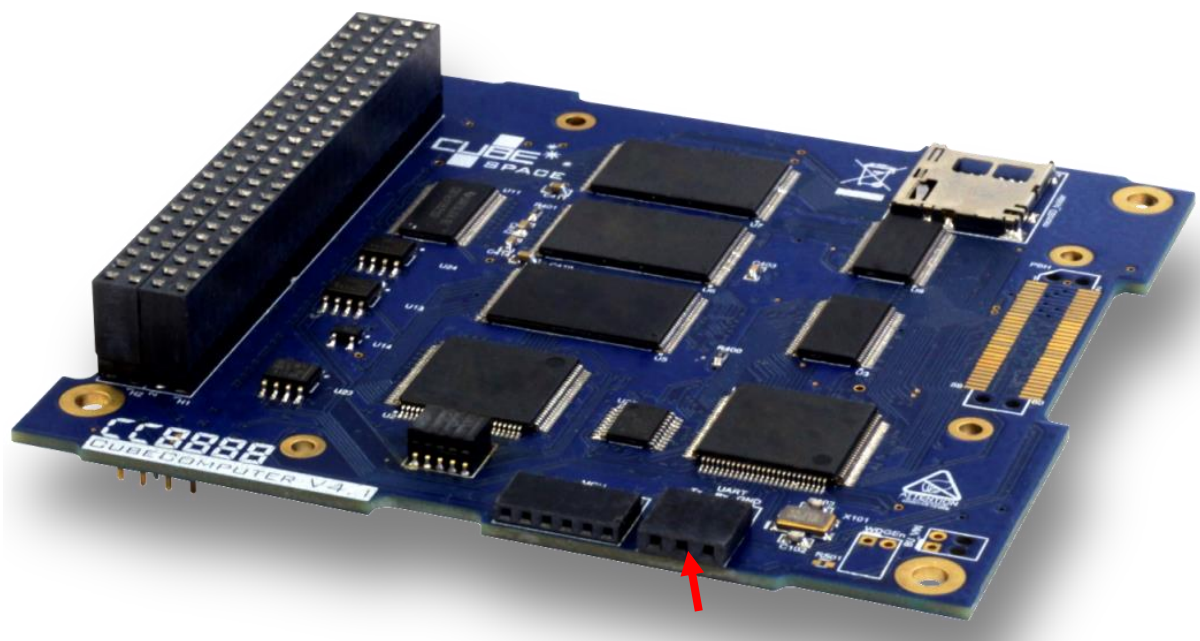


Figure 11 – CubeComputer's UART header location

7.1.1 Specifications

Table 5 – CubeComputer Physical Specifications

Specification	Value	Notes
Physical		
Mass	56 g $\pm$ 10%	Weight will vary with configuration
Dimension	90 x 96 x 10 mm	
Radiation	TID @ 20 k Rad SEE @ 60 MeV and 200 MeV	High energy particle tests have been conducted to induce SEE and to test protection circuitry and software
Electrical		
Supply Voltage	3.3 V	
Reset threshold voltage	3.0 V	Worst case. Typical value is 2.93 V and minimum is 2.88 V
Average power when running ADCS	120mW	
Peak power when running ADCS	200mW	
Environment		
Operating Temp	-10 to 70 °C	
Vibration	8 g RMS random	
Radiation	20 k Rad	

7.2 CubeSense

CubeSense is available in two different form factors: a PC104 form-factor PCB with two separate camera modules (called CubeSense V2) and a small integrated module with camera (called CubeSense V3).

7.2.1 CubeSense V2

The two camera modules on CubeSense V2 can be configured as either Nadir or Sun sensors during manufacturing. The Sun sensor has a filter that allows only intense light emitted from the sun to be detected. Therefore, camera type must be specified when placing an order. These camera modules can either be soldered directly to the CubeSense PC104 board using right-angle headers, or it can be supplied on harnesses (maximum length of 350 mm). The harness will be soldered in on the camera module side and plugged in on the PC104 module using Samtec TFM screw down headers. The camera modules should be mounted to the satellite side panels as shown in Figure 12 with the lenses protruding as far as possible so that no objects are in its FOV.

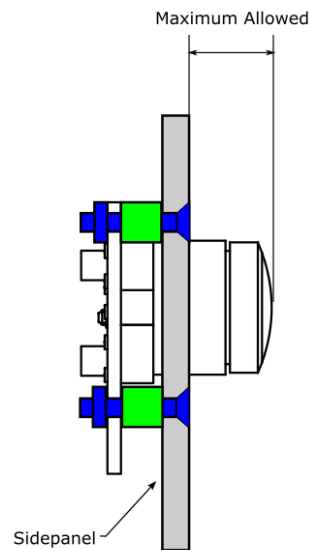


Figure 12 – Camera protruding from side panel.



Please note that for CubeSense V2, where the cameras are soldered onto the PCB, it will require removal of the side panels of the satellite structure during integration of the bundle.

The camera modules have mounting holes as shown in Figure 13. There are components populated at the back of the module, therefore all mounting should be done from the front side. The area indicated in red is a keep-out area (height = 2 mm) that is used for soldering the harness to the board. The cutout on the satellite side panel for the lens to protrude from should have a 0.5 mm larger radius to allow for deviations in lens-assembly placement.

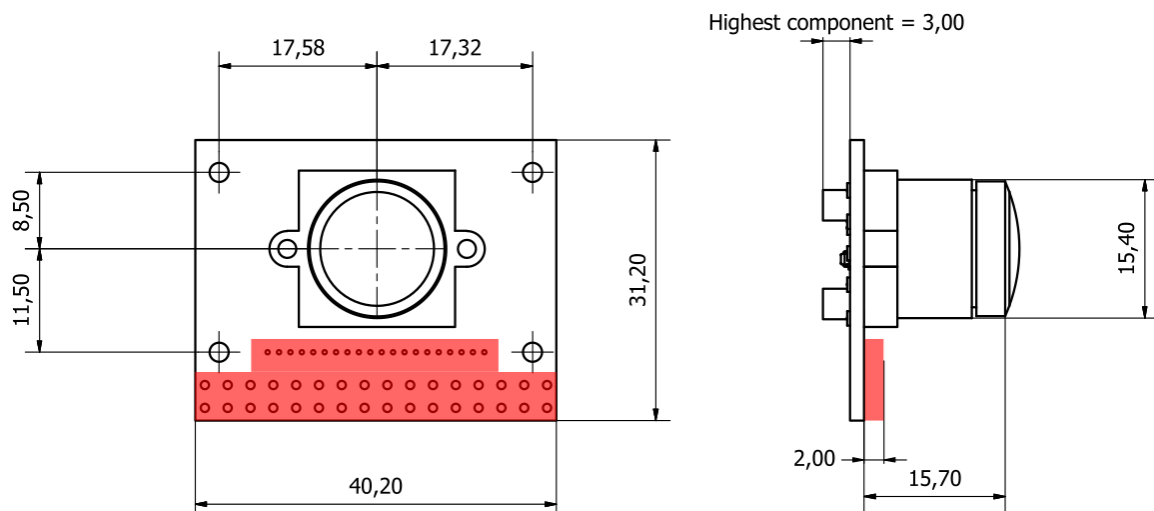


Figure 13 – Camera module of CubeSense V2.

7.2.2 CubeSense V3

The CubeSense V3 is a small camera module with integrated electronics for use in cases when space is constrained. This allows for much more compact ADCS stacks and provides the end-user the freedom to mount the cameras where they have the volume available. The CubeSense V3 only has one SRAM and one camera per module therefore has no redundancy, as the CubeSense V2 has. CubeSense V2 has an SRAM for each camera module and if one of the SRAMs fail, both cameras can still operate from the remaining SRAM.

CubeSense V3 is configured during fabrication to be either a sun sensor or a nadir sensor. The module connects to the ADCS stack with a Samtec SFSDT harness that plugs into the back of the CubeSense module. The connector for this cable can be either a straight or right-angle Samtec TFM header. The connector needs to be fixed with epoxy in place after final integration of the satellite to ensure that it does not loosen during launch. This module should be mounted to the satellite side panels with the lens again protruding as far as possible to avoid objects being visible in its FOV. The mounting hole layout is shown in Figure 14. The camera module comes with M2 threaded mounting brackets attached to the front with a hole depth of 3 mm.

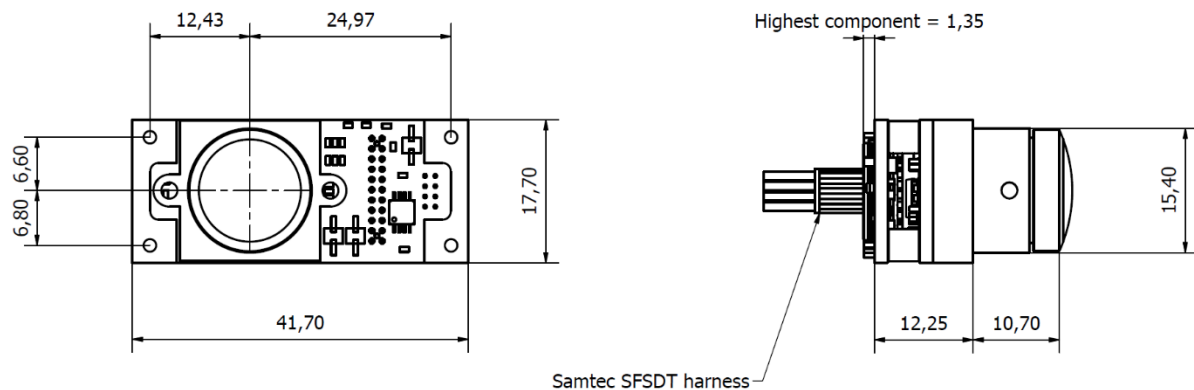


Figure 14 – CubeSense V3 dimensions.

7.2.3 Axis definition

The mounting orientation of the CubeSense cameras is specified by a set of three angles **alpha** (α), **beta** (β), and **gamma** (γ). These three angles correspond to the angles in a Euler 3-2-1 rotation sequence to rotate the sensor's coordinate frame to the SBC. The sensor coordinate frames for the CubeSense V2 and V3 are defined as follows:

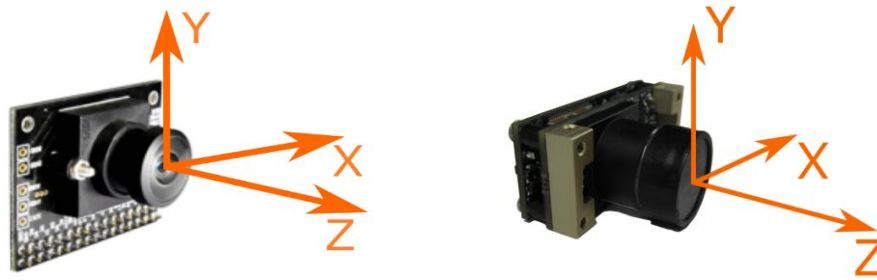


Figure 15 – Coordinate frames of CubeSense V2 camera module (left) and CubeSense V3 (right)

The rotations α , β , and γ are around the sensor's Z, Y and X axes respectively.

7.2.4 Specifications

Table 6 – CubeSense Specifications

	CubeSense V2	CubeSense V3	Notes
Physical			
Mass	80g	30g	Without harnesses
Dimensions	40.2 x 31.2 x 20 mm	41.7 x 17.7 x 26 mm	Camera modules
Electrical			
Supply Voltage	3.3 V	3.3 V	
Average Power	150 mW	100 mW	At 1Hz update rate
Peak Power	360 mW	200 mW	During image capture
Performance			
Sun sensor Accuracy	<0.2°	<0.2°	Entire FOV, 3- σ
Nadir sensor Accuracy	<0.2°	<0.2°	Entire Earth visible in FOV, 1- σ
Update Rate	1 Hz	2 Hz	
FoV	~180°	~180°	
Image size	1024x1024	1024x1024	Grayscale Images
Environmental			
Operating Temperature	-20°C to 70°C	-20°C to 70°C	
Vibration	14g RMS	14g RMS	
Radiation	24k Rad	24k Rad	

7.3 CubeTorquers (CubeRod and CubeCoil)

CubeRod is a nanosatellite magnetic torquer that uses a specially treated ferrous core with ultra-low remanence and high linearity. These rods are driven by the CubeControl. The rods come in three different sizes, depending on the application. Typically, the small CubeRods are used which are mounted on the CubeControl, whereas the medium and large sizes must be mounted separately.

CubeCoil is a low-power nanosatellite magnetic torquer with an air-coil that fits in the stack between the PCBs. It is also driven by CubeControl. In situations where stronger torques are required on the satellite a double wound coil will be used (referenced as *Double Coil*).

7.3.1 Axis definition

The axis of the torquer rods and torquer coil mounted on the CubeADCS aligns with ADCS body coordinates (ABC), where a standalone torquer (connected to CubeControl) has its positive direction away from the harness connection when a voltage is applied on the pin marked 1 and ground on pin 2. These axis definitions are shown in Figure 16 and Figure 17.

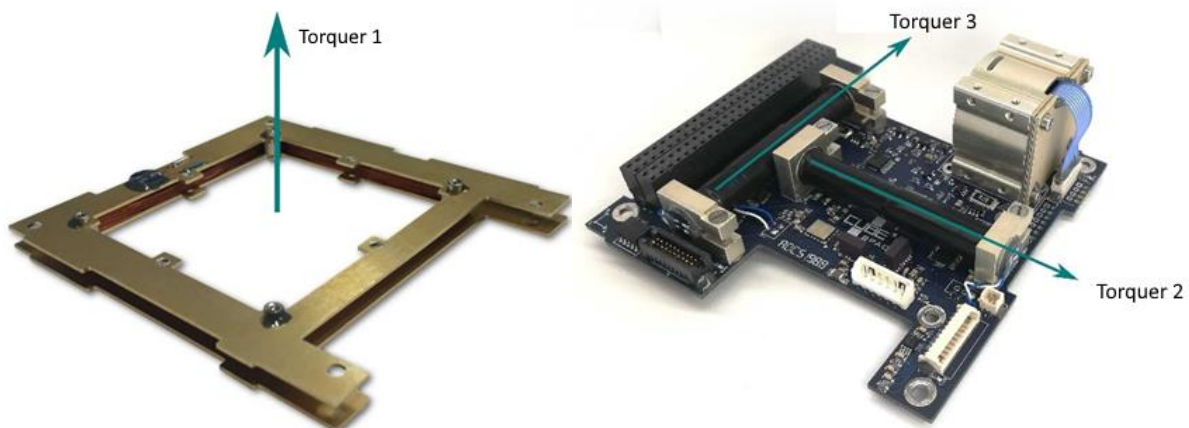


Figure 16 – Magnetorquer axis definition

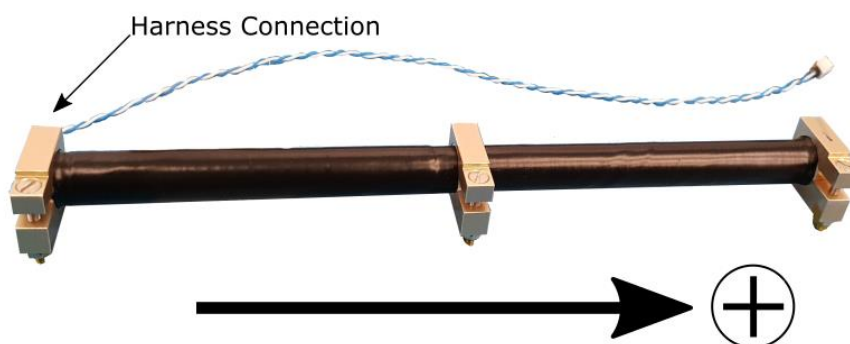


Figure 17 – Axis definition of a medium/large CubeRods

7.3.2 Specifications:

Table 7 – Magnetorquer Specifications

	Small Rod	Medium Rod	Large Rod	Coil	Double Coil
Physical					
Mass	28g	36g	72g	46g	74g
Dimensions (mm)	18x14x62	18x14x77	18x14x153	90x96x6	90x96x6
Operating Temp	-20°C to 70°C				
Vibration	14g RMS				
Electrical					
Max Continuous Current @ 25°C, 5V supply	167 mA 70 mA @ 42% PWM	80 mA	76 mA	64 mA	125 mA
Performance					
Maximum Magnetic Moment @ 25°C, 5V supply	±0.48 Am ² ±0.2 Am ² @ 42% PWM	±0.61 Am ²	±1.88 Am ²	±0.13 Am ²	±0.27 Am ²
Environmental					
Operating Temp	-20°C to 70°C				
Vibration	14g RMS				
Outgassing of heat shrink	TML = 0.36% CVCM = 0.04%				
End Caps	Alodine Aluminium				

7.4 CubeControl

CubeControl is an actuator and sensor interface module. It implements magnetic control actuation by reading up to two magnetometers and driving three magnetorquer rods. It also controls a momentum wheel, and interfaces with ten coarse sun sensors, and three MEMS rate sensors.

7.4.1 Orientation and Axis definition

The rate sensors align with the X/Y/Z axes of the ADCS Body Coordinates. If a momentum wheel is mounted on CubeControl, it can be aligned to any axis.

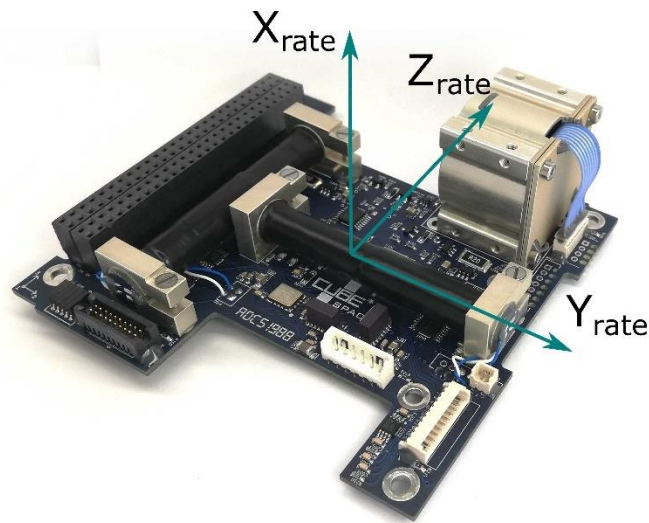


Figure 18 – Rate sensor axis definition

7.4.2 Specifications

The CubeControl and its subsystems' specifications are shown below in Table 8.

Table 8 – CubeControl Specifications

Specification	Value	Notes
Physical		
Dimension (mm)	90 x 96 x 19.5	Without momentum wheel
	90 x 96 x 34	With momentum wheel
Mass		
Total mass	145 g	Including: PCB, momentum wheel and 2 magnetorquer rods
PCB and components	44 g	
Momentum wheel	47 g	
Magnetorquer rods	54 g	Y-axis and Z-axis rods
Electrical		
Supply Voltage	3.3V & 5V	
	Vbat: 6 to 16V	For magnetometer deployment. Voltage must be specified during ordering.
	Vbat: 6.5V to 16V	For momentum wheel Turn-on threshold: 7.4 V Turn-off threshold: 6.5 V
Average power	200mW	No Actuators
Momentum wheel power	85mW	Holding at 2000 rpm
	240mW	Max acceleration
Performance		
Measurement noise	< 0.015 °/s	1 σ
Update rate	1 Hz	Averaged output
Angular Random Walk	0.28°/ $\sqrt{h}$	

Bias instability	24°/h	
Momentum wheel (if included)		
Maximum momentum storage	1.7 mNms	@ 8000 rpm
Maximum wheel speed	± 8000 rpm	
Maximum torque	0.15 mNm	
Wheel inertia	2.0 kg.mm ²	
Environment		
Operating Temp	-10 to 70 °C	
Vibration	8 g RMS random	
Radiation	24k Rad	

7.5 CubeMag

7.5.1 External deployable magnetometer

The primary magnetometer is housed on a small PCB, enclosed by an aluminum housing. The bare magnetometer housing measures 16.8 mm x 16.4 mm x 6.3 mm, as shown in Figure 19. The tolerance on dimensions are ± 0.3 mm.

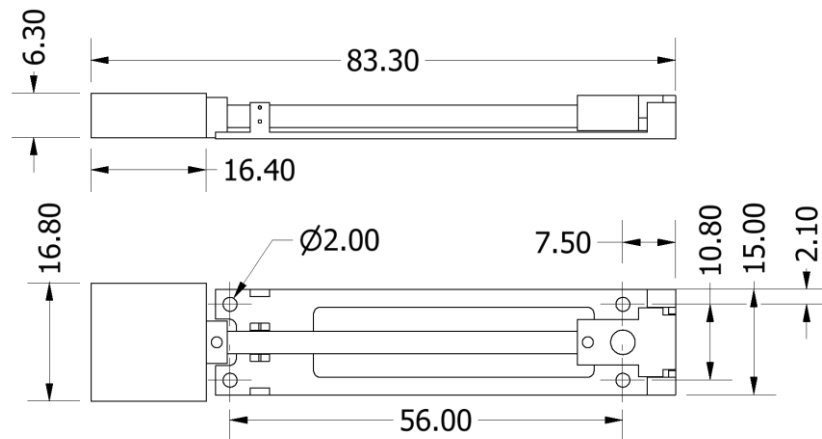


Figure 19 – Deployable magnetometer dimensions.

The recommended mounting pattern is shown in Figure 20. Four M2 screws (A4 steel) should be used to mount the magnetometer. The 5mm diameter hole allows the connector to pass through the satellite body.

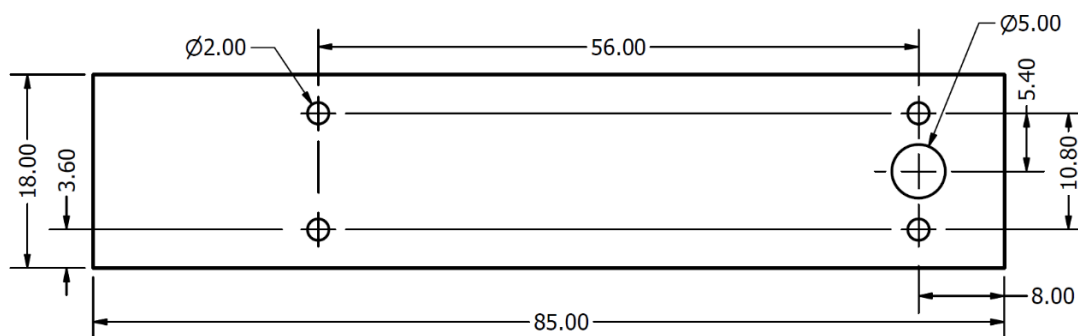


Figure 20 – Deployable magnetometer recommended mounting pattern.

The magnetometer cable **must not** be routed inside the satellite a manner such that the cable **is tight** where it exits from the magnetometer. **The cable must exit perpendicular to the mounting facet for 1 cm, it should have 2 cm slack and must be loose (movable with slight touch).** See **Figure 21** for the correct illustration. **Failure to conform may influence magnetometer deployment significantly.**



Figure 21 – Deployable magnetometer recommended mounting pattern.

The magnetometer is supplied with a deployable boom. The boom separates the magnetometer 8 cm from the satellite body, limiting the effect of electromagnetic interference and lowering the measurement noise. The magnetometer should not be placed in close proximity of any other part of the satellite that causes significant disturbances, like electrical motors or antennas. Although the magnetometer can be supplied without the deployment mechanism, it is strongly recommended that the magnetometer is deployable. This is owing to the magnetometer's important role during detumbling.

The deployable boom consists of a rod with the magnetometer in its enclosure at one end and a mounting bracket with a spring on the other, as illustrated in Figure 22. The bracket should be mounted on the outside of the satellite structure. The magnetometer will be supplied in the stowed position; deployment can be done in-flight via the appropriate TC.



The deployable magnetometer can NOT be refurbished; therefore, a test deployment should not be done on the ground.



Figure 22 – Deployable magnetometer stowed (left) and deployed (right).

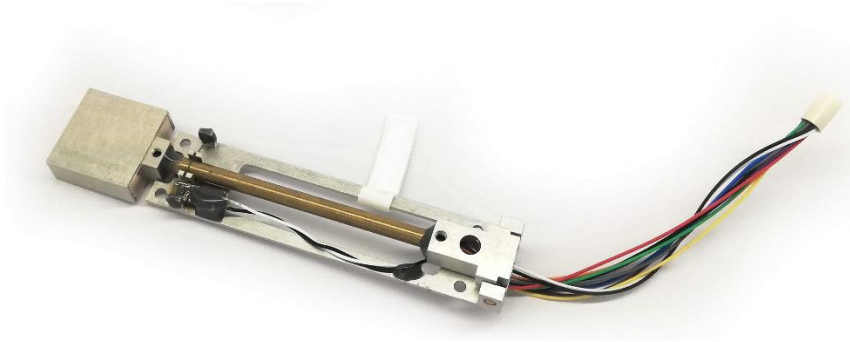


Figure 23 – Deployable magnetometer stowed.

The magnetometer interfaces with the CubeControl module using an Omnetics Nano Circular 11-way in-line connector set. The harness is terminated in an 11-way Molex PicoBlade female connector. An illustration of the magnetometer connection can be seen in Figure 24 and Figure 25.

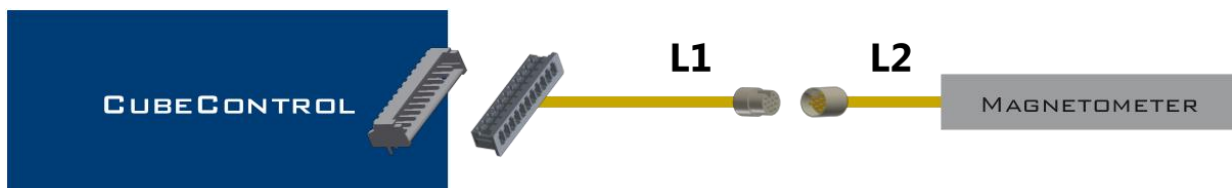


Figure 24 – Magnetometer wiring diagram.

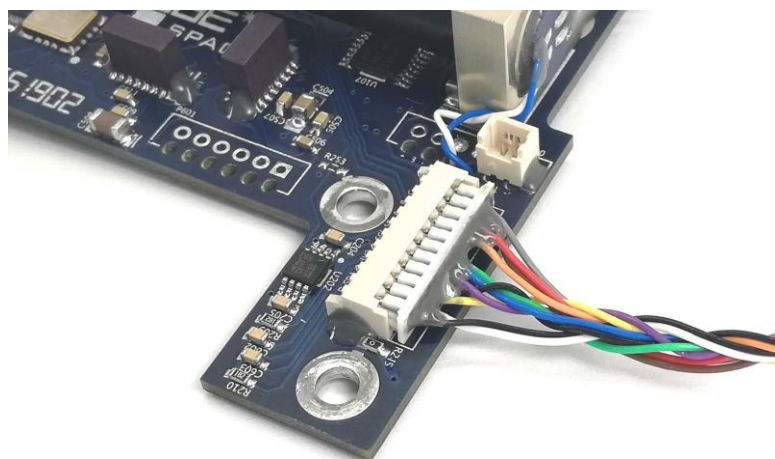


Figure 25 – Deployable magnetometer CubeControl header

The length between the CubeControl and the in-line connection can be configured to the user's requirements. The maximum total length of the harness is 350 mm. The in-line Omnetics connector has a diameter of 3.9 mm.

7.5.2 Magnetometer deployment current

When the "deploy magnetometer" command is given, the nichrome burn circuit is activated for 5s. It will draw 0.62A for a battery supply voltage of 8V. The deployable magnetometer CANNOT be refurbished; therefore, a test deployment should not be done on the ground.

7.5.3 Redundant magnetometer

The CubeADCS unit offers an interface to an optional redundant magnetometer. The magnetometer is housed in the same enclosure as the primary magnetometer, but it is not deployable. The dimensions of the redundant magnetometer is shown below in Figure 26.

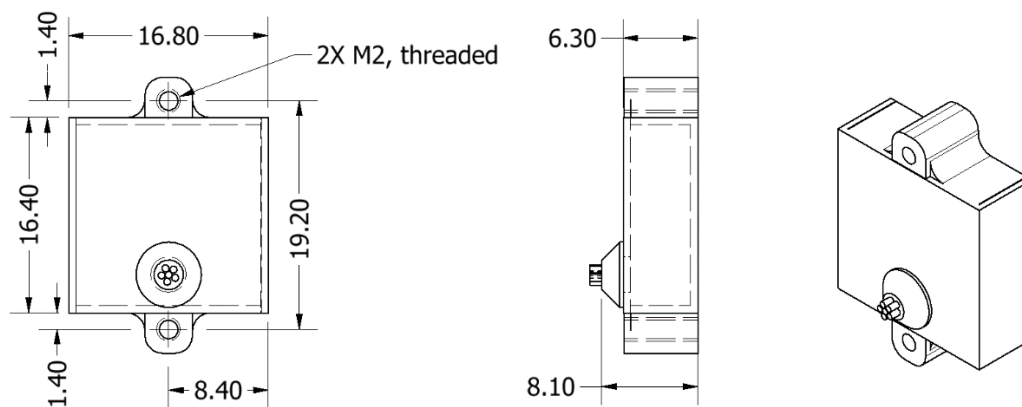


Figure 26 – Redundant magnetometer dimensions.



Figure 27 Redundant Magnetometer

The redundant magnetometer is connected to the CubeControl module via a 6-way harness (not in-line) and the maximum harness length is 350 mm.

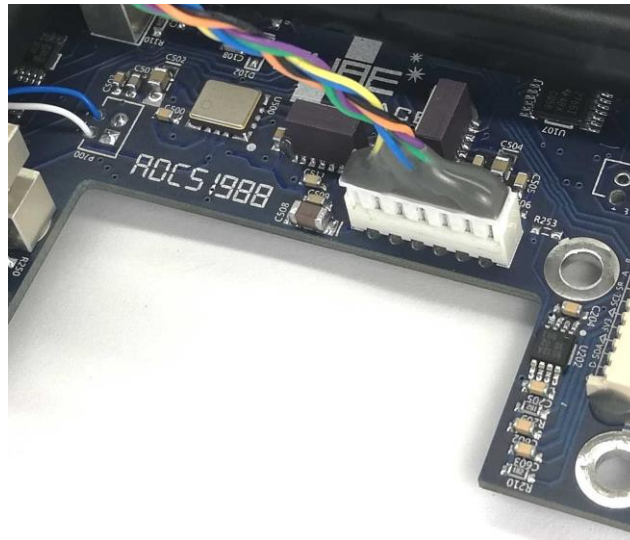


Figure 28 Redundant Magnetometer connection on CubeControl

7.5.4 Axis definition

The mounting of the magnetometer is specified by a set of three angles **alpha (α)**, **beta (β)**, and **gamma (γ)**. These three angles correspond to the angles in a Euler 3-2-1 rotation sequence to rotate the sensor's coordinate frame to the SBC. The sensor coordinate frames for the various sensors are defined as follows:

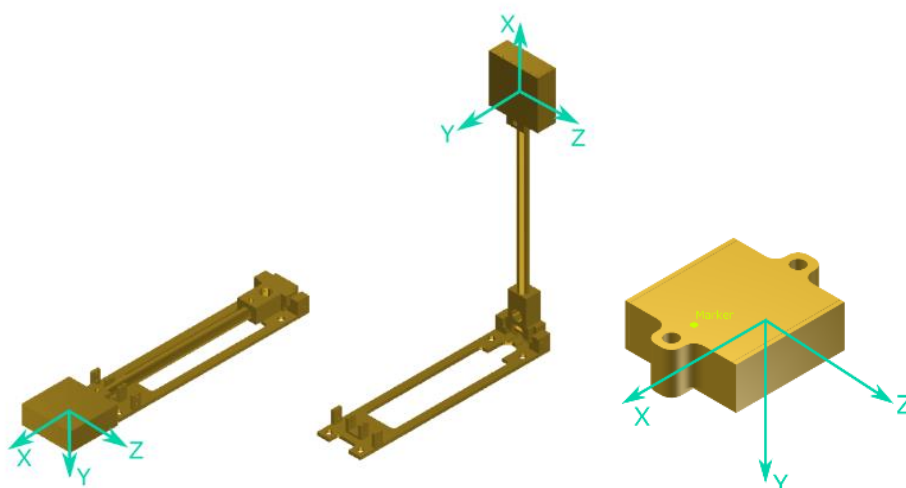


Figure 29 CubeMag Axis definition

The rotations α , β , and γ are around the sensor's Z, Y and X axes respectively.

7.5.5 Specifications

Table 9 – CubeMag Specifications

Specification	CubeMag	Redundant	Notes
Physical			
Mass	9.1 g	6.2 g	
Dimension	83.3 x 17.2 x 6.7 mm	23.4 x 16.8 x 8.1 mm	
Operating Temp	-20 to 85 °C		
Vibration	8 g RMS random		
Radiation	24k Rad		
Electrical			
Supply Voltage	3.3 V		
Average power	35 mW		
Performance			
Measurement noise	< 50 nT	< 50 nT	1σ (per X/Y/Z channel)
Update rate	1 Hz	1 Hz	Averaged output

7.6 Coarse Sun sensors

The CubeADCS unit can interface with up to ten coarse Sun sensor (CSS) photodiodes, each mounted on a small PCB. The dimensions of each CSS PCB are 3.8 mm x 10.8 mm, and the height is 1.7 mm.

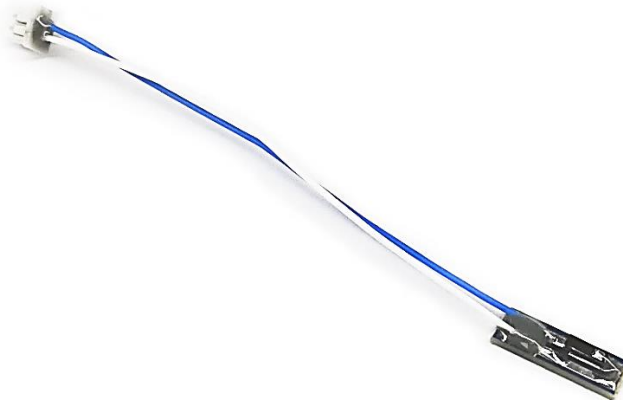


Figure 30 – Single coarse Sun sensor PCB and cable

The ten CSSs are each connected to CubeControl by a Molex PicoBlade 2-way in-line connector to ease the satellite integration process. The ten 2-way harnesses are terminated in a single 20-way female connector, of which the male connector is populated on CubeControl. Figure 31 illustrates the above-mentioned wiring of the CSS units. The PCBs with the CSSs are connected to a short stub that is 50mm long. The in-line harness can be configured to be up to 350mm, with 300mm being the default length.

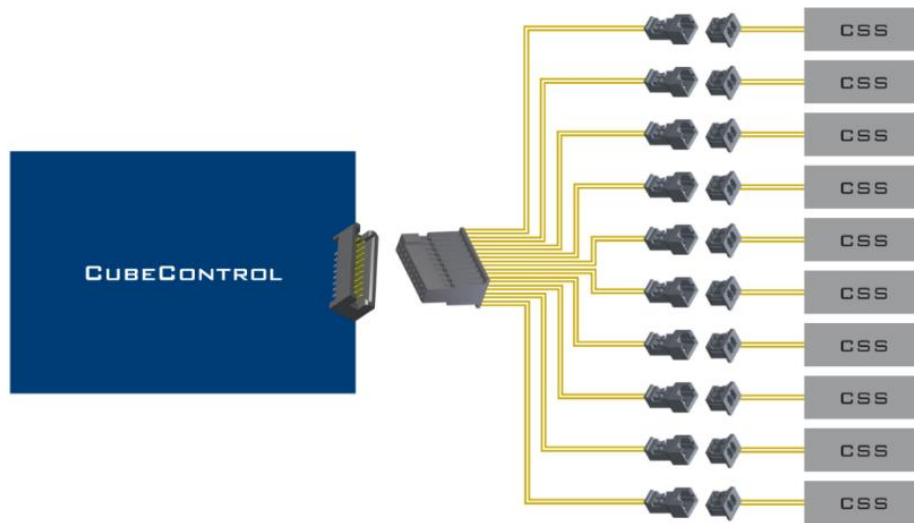


Figure 31 – Coarse Sun sensor wiring diagram.

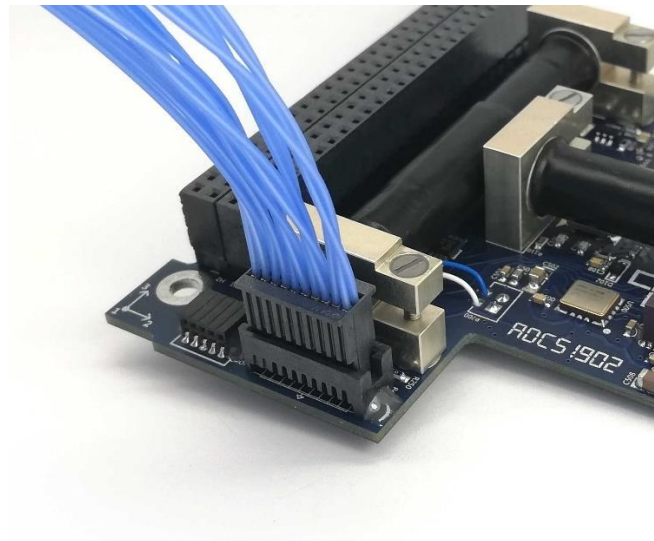


Figure 32 – Location of CSS header

The CSSs do not have mounting holes – they should be attached to the satellite body using epoxy as shown in Figure 33. At least six photodiodes, each on a different facet of the satellite, are required to obtain a coarse Sun vector throughout the entire sunlit part of the orbit. The four additional sensors can be used for redundancy or in cases where sensors are shadowed or obstructed.

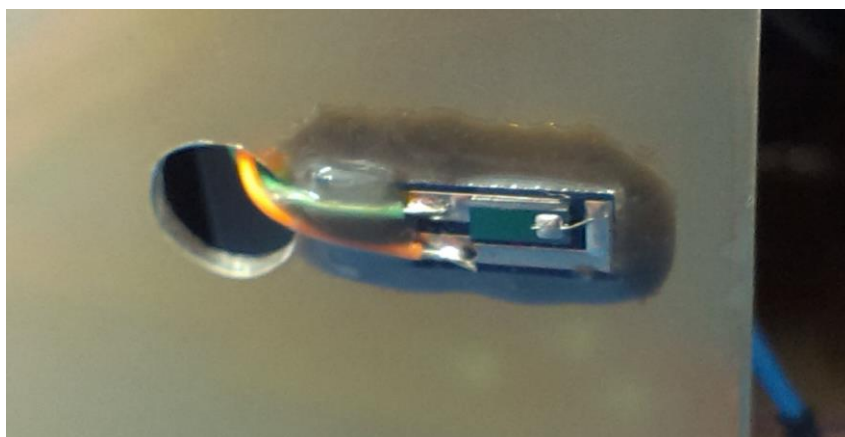


Figure 33 – Coarse Sun sensor epoxied to satellite body.

7.6.1 Specifications

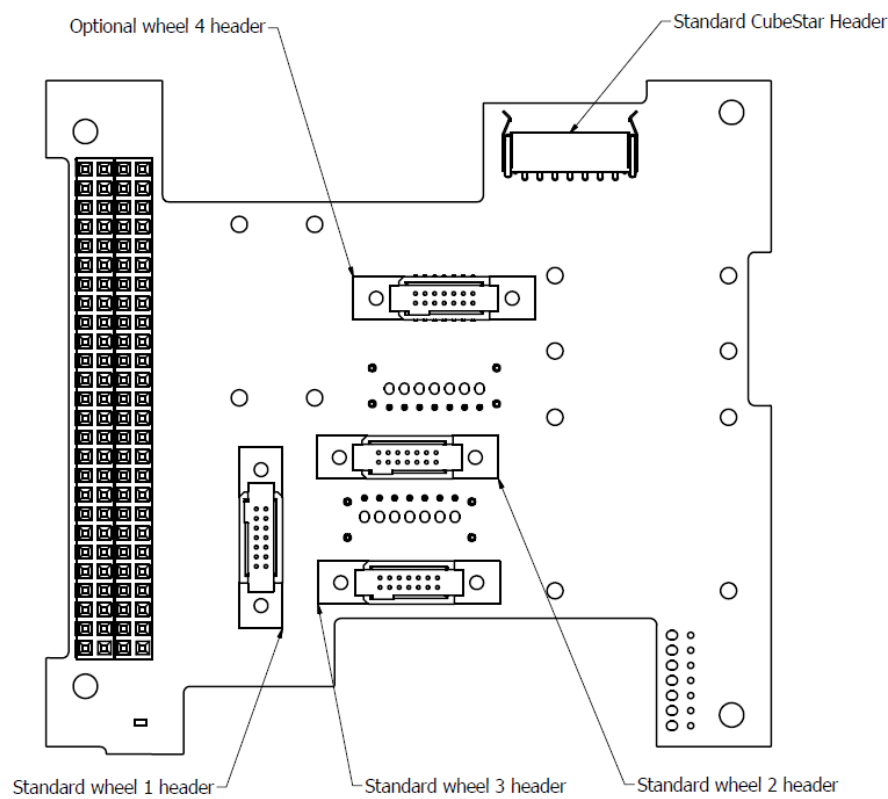
Specification	Value	Notes
Physical		
Mass	0.3 g	PCB and 50mm connector stub
Dimension	3.8 x 10.8 x 1.7 mm	
Electrical		
Average power	< 5 mW	All sensors
Performance		
Measurement accuracy	< 10°	1 σ
Update rate	1 Hz	
Environmental		
Operating Temp	-40 to 125 °C	
Vibration	8 g RMS random	
Radiation	24k Rad	

7.7 CubeConnect

The CubeConnect board acts as an interface board for several of the loose peripheral components to connect to the PC104 stack. This board has headers for connecting CubeWheels, CubeStar(s) and the CubeSense V3 modules. The CubeConnect is available in two different versions: A PC104 form-factor board with cutouts as seen in Figure 34, and the smaller version called CubeConnect Micro as seen in Figure 37.

7.7.1 CubeConnect

The board has 3 Harwin M80 connectors that can be used for either a CubeStar module or a CubeSense module. The board also has 4 Samtec TFM screw down headers that can be used for the wheels. The locations of these headers are shown in Figure 34.



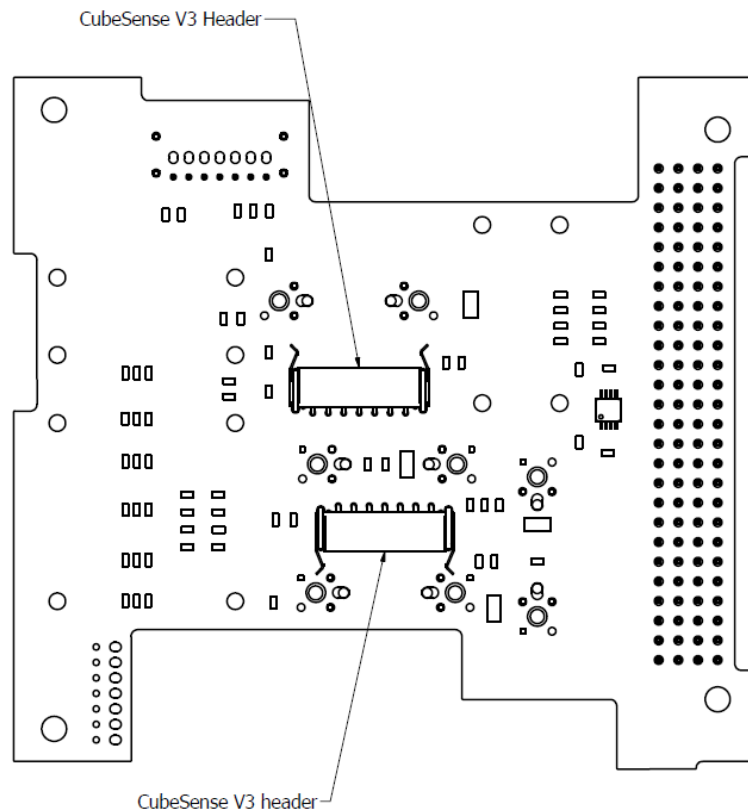


Figure 34 – CubeConnect interface board.

The M80 headers can be forgone to rather have the CubeStar/CubeSense harnesses soldered in with strain relief as seen in Figure 35.

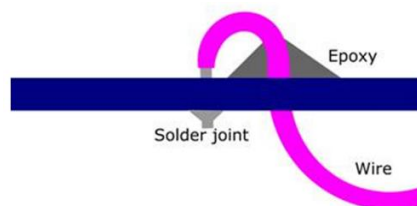


Figure 35 – Strain relief of soldered harnesses.

For modules that have small wheels, they can be mounted directly on the CubeConnect board as seen in the CubeADCS drawings in Section Standard CubeADCS solutions.

7.7.2 CubeConnect Micro

For bundles with loose wheels, the CubeConnect Micro can be used which allows for a very compact CubeADCS stack as seen in Figure 36. This CubeConnect only has the option to solder in the harnesses and can only handle 3 wheels, 2 CubeSense modules and 1 CubeStar. The location where these harnesses can be soldered in is fixed and careful provision should

be made for routing these harnesses to the associated sensor/actuator. The location of the harness solder points is shown in Figure 37.

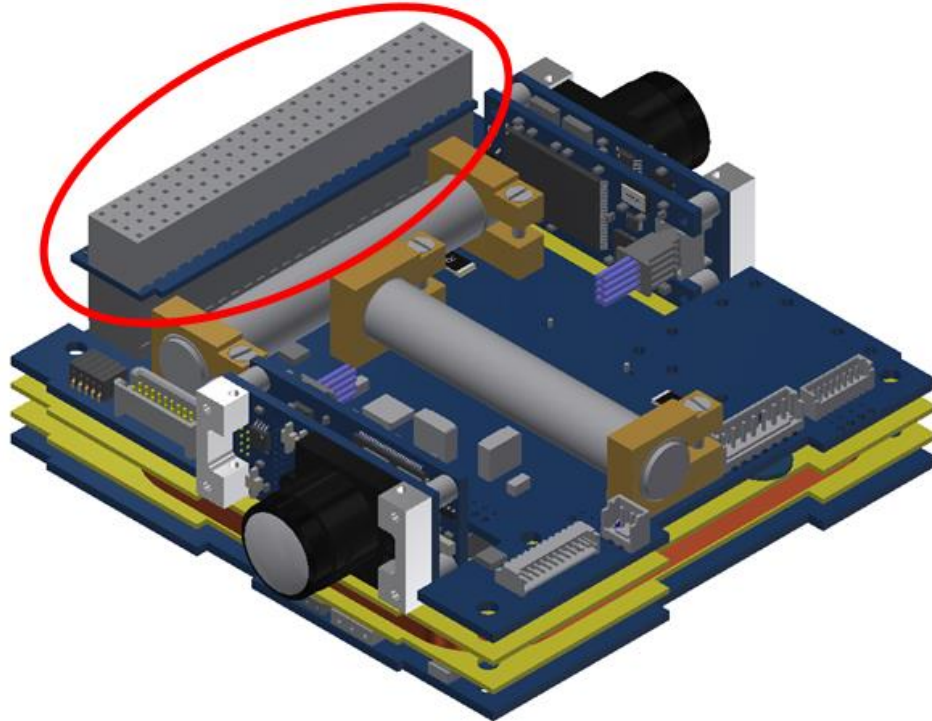


Figure 36 – Micro CubeConnect on ADCS stack.

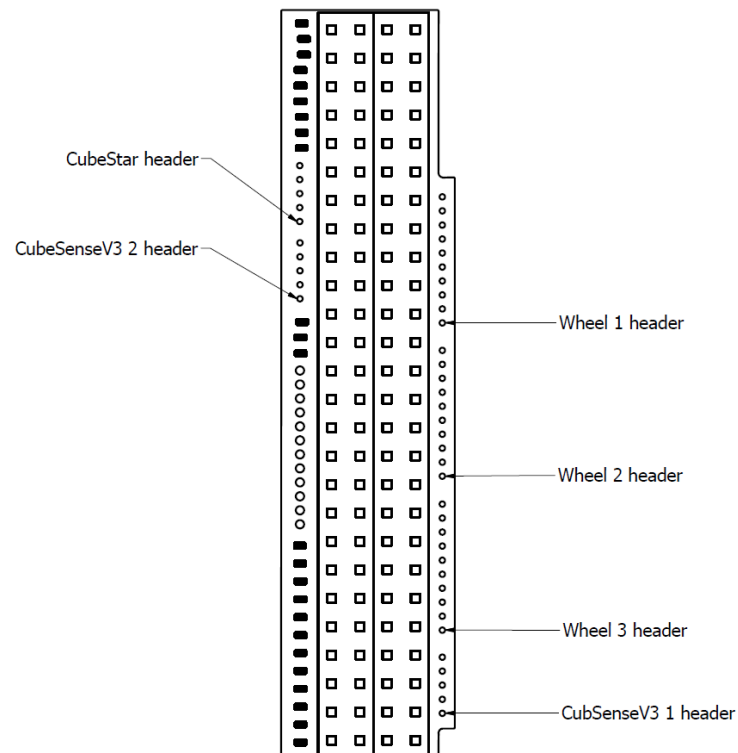


Figure 37 – Micro CubeConnect.

7.8 CubeWheel

The mounting of the three CubeWheel reaction wheels will depend on the size of the CubeWheels. In the case of small CubeWheels, the wheels can be: (1) mounted on the regular CubeConnect and integrated onto the main ADCS PC104 stack; (2) mounted on the regular CubeConnect and supplied separate from the ADCS stack; or (3) supplied loose with a CubeConnect Micro in the ADCS stack. In this case, the user must mount the wheels to their satellite frame via a bracket. In the case of medium or large CubeWheels, there is only the option where the user must mount the wheels. The harnesses that connect the CubeADCS to the wheels should have adequate length to cover the required distance with some added slack to avoid strain on the connectors during integration. The disc inside a CubeWheel has been balanced to avoid jitter when the wheel is spinning, but it is still good practise to not mount any CubeWheel on the same mounting surface of sensitive optics.

7.8.1 Small CubeWheel

The small CubeWheel has three sets of four M2 mounting holes on three different facets, as illustrated in Figure 38. The outer dimensions of the small CubeWheel are 28 mm x 28 mm x 26.1 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses.

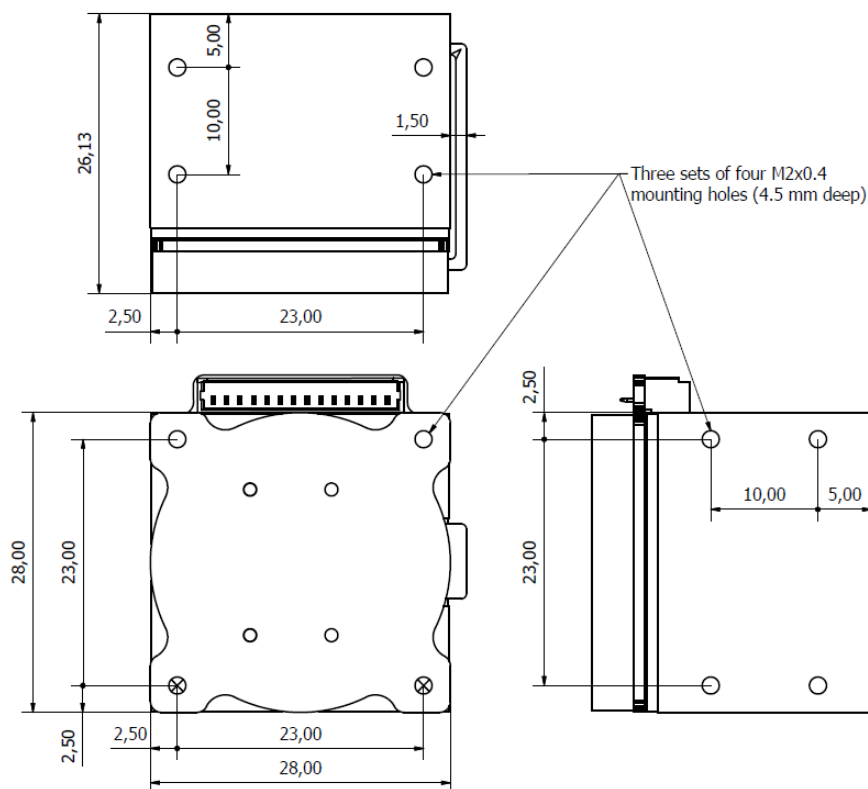


Figure 38 – Small CubeWheel mechanical interface.

7.8.2 Small Plus CubeWheel

The Small Plus CubeWheel has three sets of four M2 mounting holes on three different facets, as illustrated in Figure 39. The outer dimensions of the small CubeWheel are 33.4 mm x 33.4 mm x 31 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses.

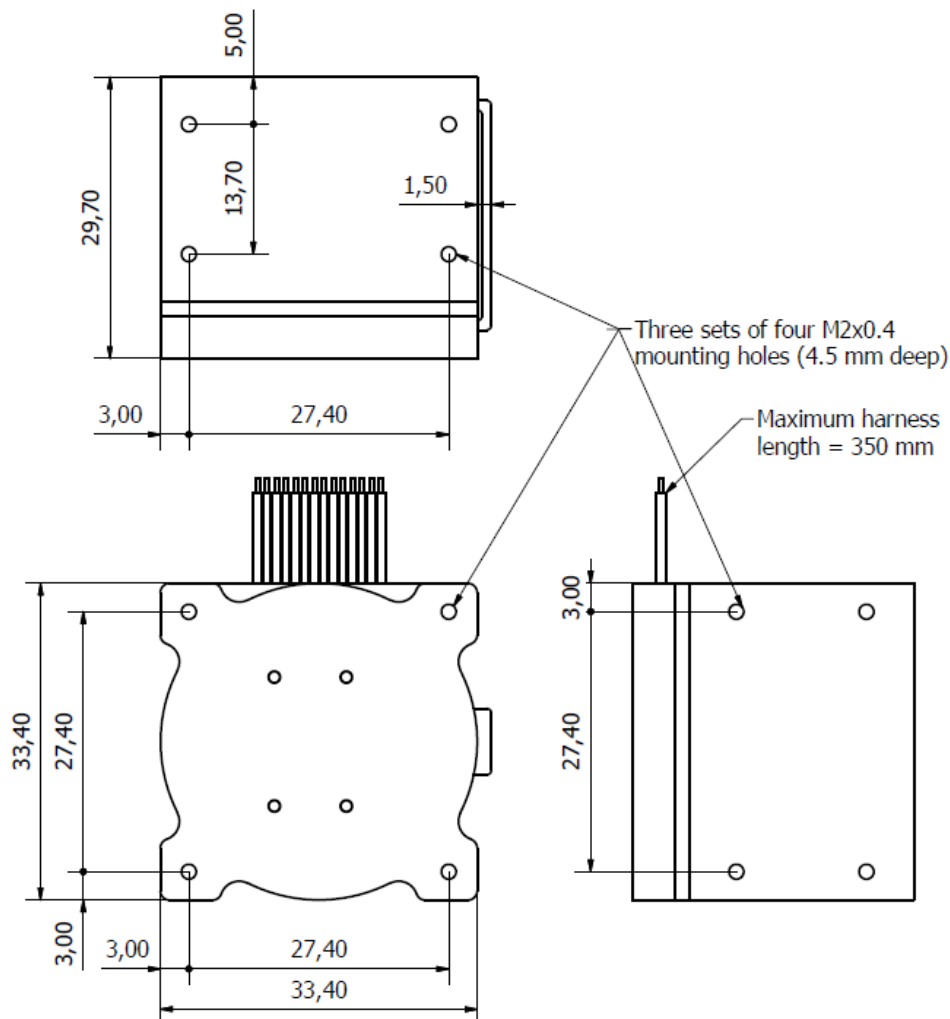


Figure 39 – Small Plus CubeWheel mechanical interface.

7.8.3 Medium CubeWheel

The medium CubeWheel has three sets of four M3 mounting holes on three different facets, as illustrated in Figure 40. The outer dimensions of the medium CubeWheel are 46 mm x 46 mm x 31.5 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses. The M3 mounting holes are fitted with A2 (304) stainless steel heli-coils to provide a stronger, more durable thread.

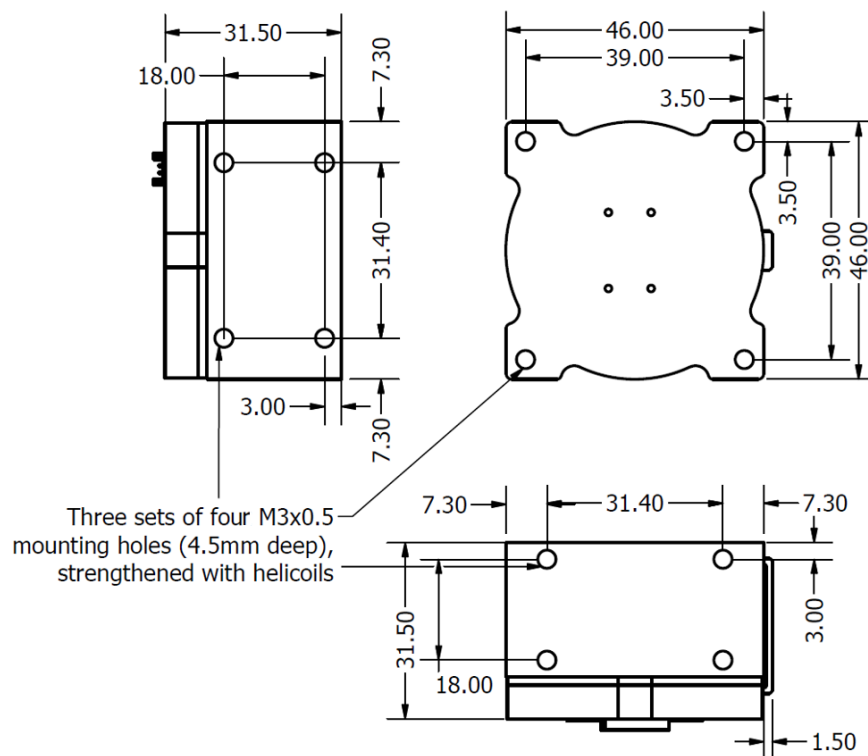


Figure 40 – Medium CubeWheel mechanical interface.

7.8.4 Large CubeWheel

The large CubeWheel unit has three sets of four M3 mounting holes on three different facets, as illustrated in Figure 41. The outer dimensions of the large CubeWheel are 57 mm x 57 mm x 31.5 mm. Note that an additional 1.5 mm is required on one side of the CubeWheel for internal harnesses. The M3 mounting holes are fitted with A2 (304) stainless steel heli-coils to provide a stronger, more durable thread.

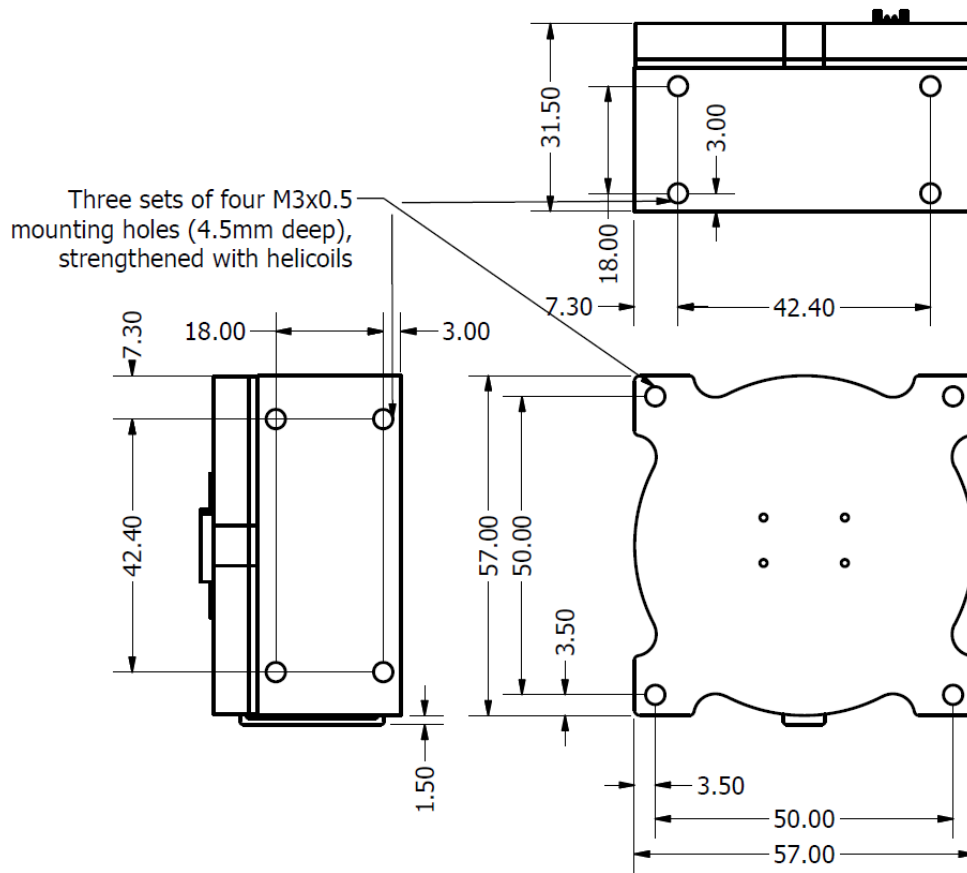
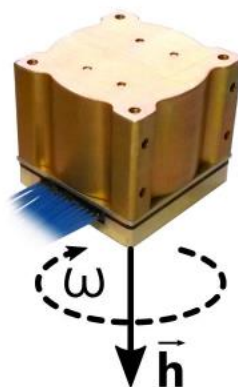


Figure 41 – Large CubeWheel mechanical interface.

7.8.5 Axis definition

The reaction wheels are either mounted on a PCB as part of the CubeADCS stack, or mounted separately by the client. In either case, the reaction/momentum wheel's positive direction is defined as:



If the user mounts the wheels, care must be taken to determine the mounting direction of the wheel according to the definition above.

7.8.6 Specifications

The CubeWheels have a turn-on threshold of 6.5 V, but lower voltages are tolerated without damage to the units. The maximum tested bus voltage is 16.0 V. For the wheels to perform optimally, a battery voltage of at least 8V is recommended.

Specification CubeWheel	Small	Small plus	Medium	Large	Notes
Physical					
Mass	60 g	90 g	150 g	225 g	Without harness
Electrical					
Supply Voltage	3V3 & Vbat	3V3 & Vbat	3V3 & Vbat	3V3 & Vbat	
Idle Power	65 mW	65 mW	65 mW	65 mW	@ 0 RPM
Average power	150 mW	190 mW	190 mW	350 mW	@ 2000 RPM
Peak power	0.65 W	2.3 W	2.3 W	4.5 W	@ max torque
Performance					
Maximum momentum storage	1.7 mNms	3.6 mNms	10.8 mNms	30.0 mNms	
Maximum wheel speed	± 8000 rpm	± 6000 rpm	± 6000 rpm	± 6000 rpm	
Maximum torque	0.23 mNm	1.0 mNm	1.0 mNm	2.3 mNm	
Static Imbalance	<0.003 g·cm	<0.004 g·cm	<0.004 g·cm	<0.006 g·cm	
Dynamic Imbalance	<0.005 g·cm ²	<0.014 g·cm ²	<0.014 g·cm ²	<0.05 g·cm ²	
Enviromental					
Operating Temp	-10 to 60 °C	-10 to 60 °C	-10 to 60 °C	-10 to 60 °C	
Vibration	14 g RMS	8.9 g RMS	8.9 g RMS	8.9 g RMS	
Radiation	24 kRad	24 kRad	24 kRad	24 kRad	

7.9 CubeStar

CubeStar is a miniature star tracker specifically intended for, but not limited to usage in low-power, performance-critical CubeSat missions.

7.9.1 Interface Header

Power and communication lines are connected through the *HARWIN M80-8760722 L-Tek* single inline male interface header.

The position of the CubeStar interface header is shown in Figure 42.



Figure 42 – CubeStar Interface Header Location

Figure 42 shows a drawing of the interface header with pin one, the first pin on the left, indicated with a designator. On CubeStar, pin one of the interface header can be identified as the closest pin to the stand-off post on the edge of the PCB. Pin seven is the pin closest to the center of the PCB.

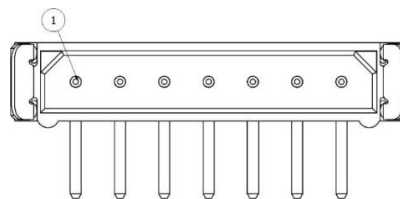


Figure 43 – CubeStar Interface Header

7.9.2 Physical Specifications

The total volume of CubeStar is 50 mm x 35 mm x 54.78 mm, with a tolerance of ± 1 mm added on the CubeStar height to account for the calibrated lens position.

The outer dimensions of the CubeStar hardware are shown in Figure 44. Each circuit board in the PCB stack has dimensions of 50 mm x 35 mm x 1.61 mm. In this figure, the total height of CubeStar, measured from the highest component on the bottom of the first PCB to the end of the lens is 54.78 mm.

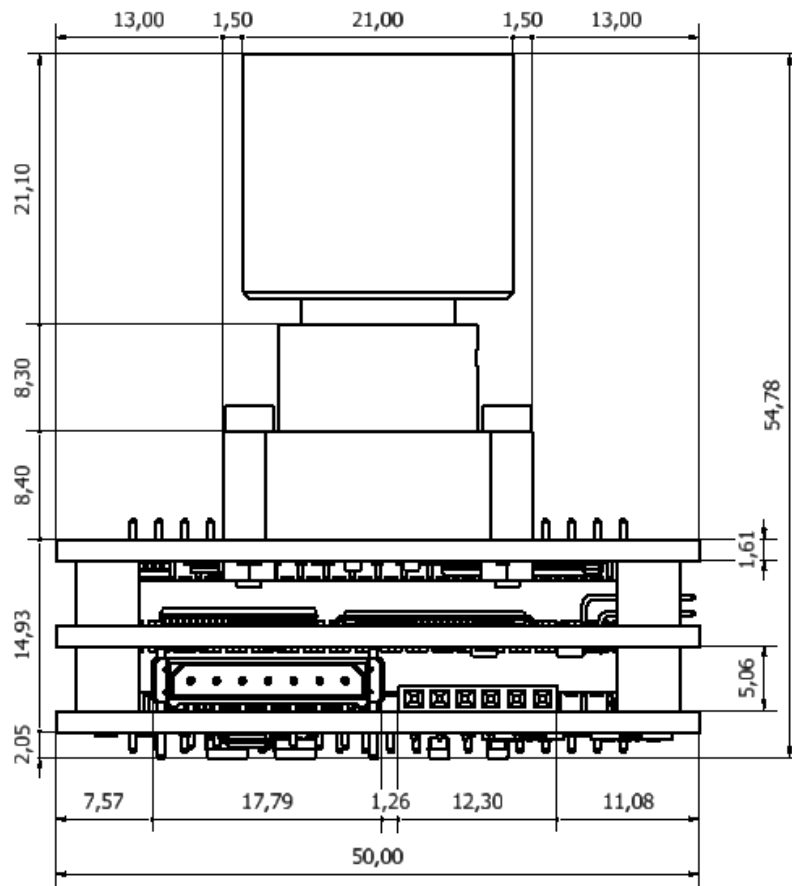


Figure 44 – CubeStar Side View with Dimensions

It should be noted that the interface header on CubeStar has latches that hold the mating connector in place. These latches protrude over the PCB by 1.61 mm. A front view of the CubeStar hardware, illustrating this latch overhang is shown in Figure 45. The size of the mating connector and wires must be taken into consideration when planning the placement of CubeStar in the satellite.

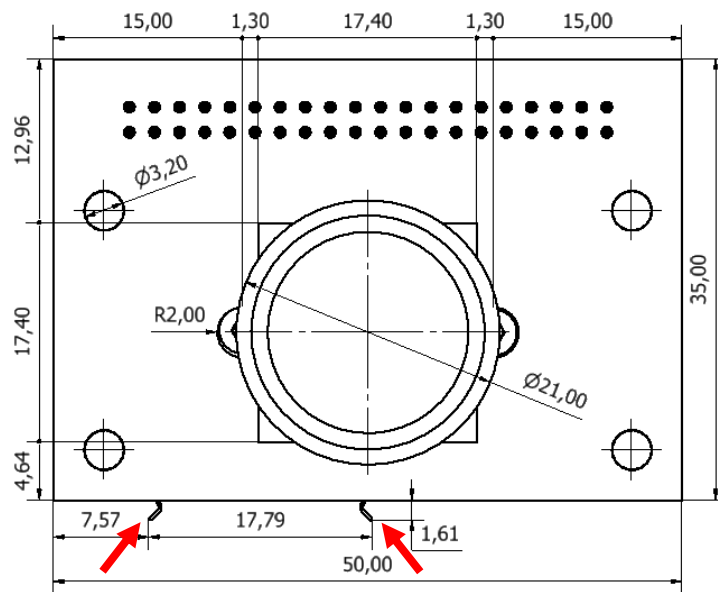


Figure 45 – CubeStar Front View with red arrows pointing at connector latch

7.9.3 Field of View

The lens used on CubeStar has a horizontal FoV of 58° and a vertical FoV of 47°. As this lens has a large angle of view, care should be taken to ensure that there are no obstructions in, or near the lens view cone. Internally, the FoV is artificially limited to 42° by using a reduced region of interest of 937 x 937 pixels. An illustration of this FoV reduction technique is given in Figure 46. In this image, the outer rectangle shows the image sensor bounds, the large circle the total unshaded view cone of the lens, and the shaded area the square 42° region of interest.

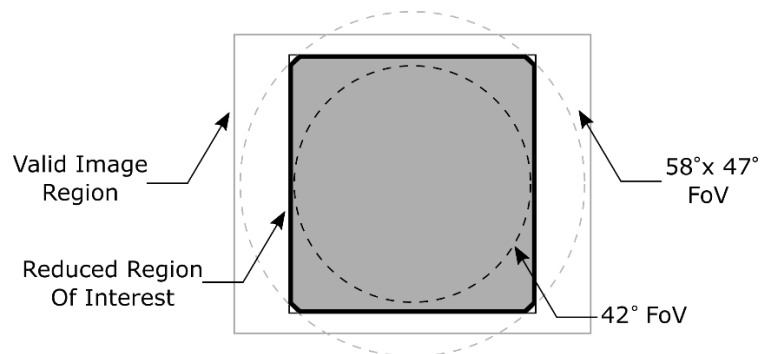


Figure 46 – CubeStar Valid Image Region

The star tracker view-cone is shown in Figure 47. The successful operation of the star tracker cannot be guaranteed if this view-cone is obstructed by deployable satellite members.

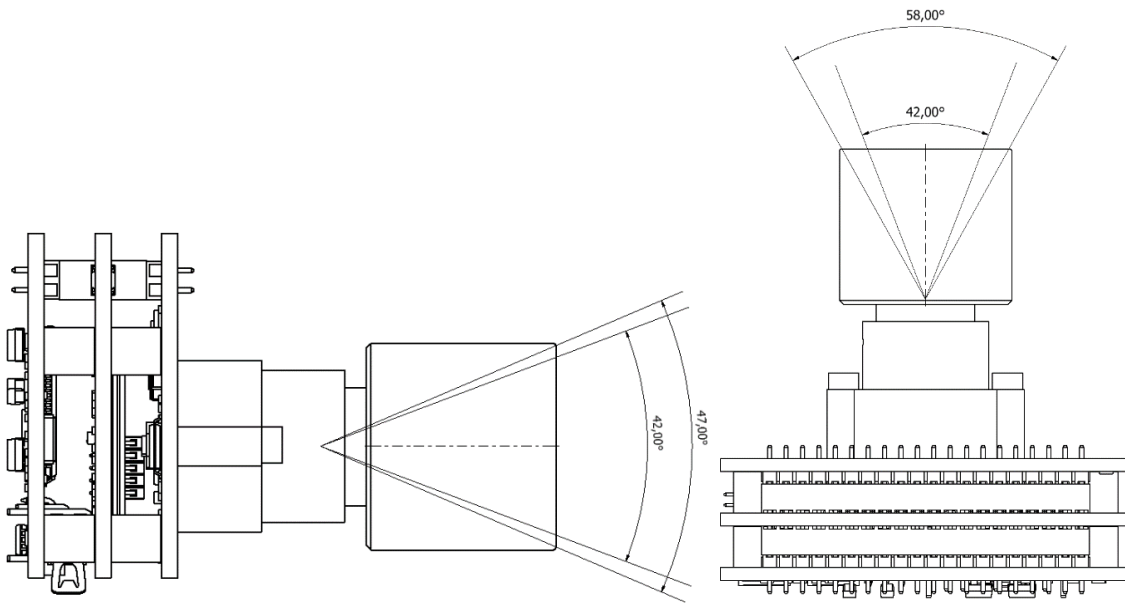


Figure 47 – CubeStar Field of View

7.9.4 Mounting Considerations

The four mounting holes each have a 3.2 mm diameter to accommodate M3 sized bolts. Figure 48 shows the spacing of the mounting holes. CubeStar is supplied with four A4 grade stainless steel bolts. If required, these bolts can be used to mount CubeStar. It is important to ensure that spacing of at least 3mm is left between the bottom PCB and the rest of the satellite when mounting CubeStar. This spacing is required to prevent electrical shorts.

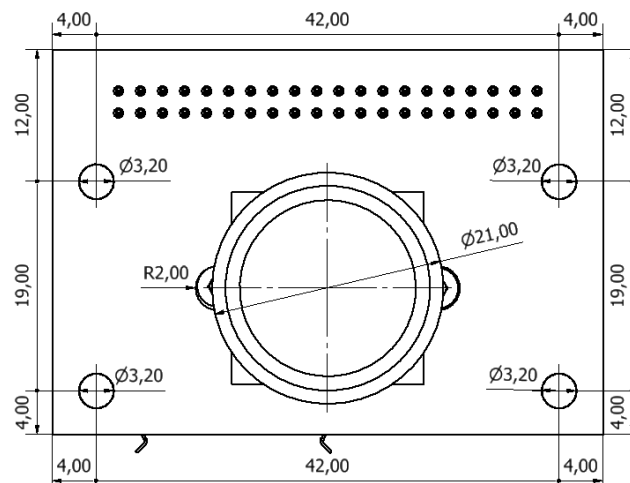


Figure 48 – CubeStar Mounting Holes

7.9.5 Lens Position

The lens is made up of multiple glass elements that can be damaged if not adequately supported. The lens holder and lens position are shown in Figure 49.

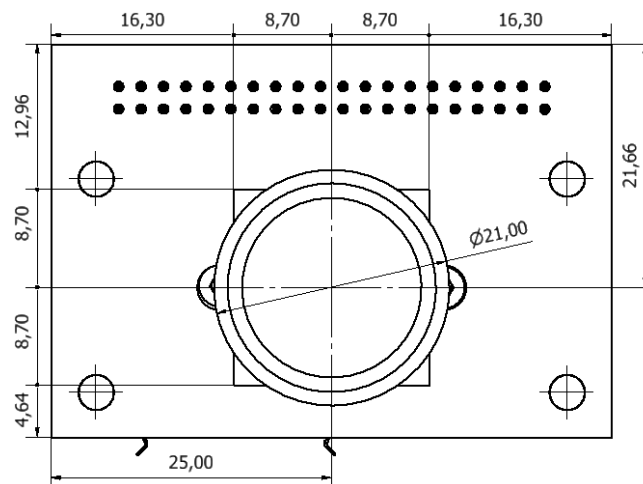


Figure 49 – CubeStar Lens Position

7.9.6 Mounting Restrictions

There are several important factors to consider before choosing a location to mount CubeStar. Although these mounting suggestions are only given as guidelines, they are advised for consideration to ensure optimal star tracker operation.

To ensure optimal performance, no panels, deployable structures, or objects should be mounted in, or close to the star tracker FoV ($58^\circ \times 47^\circ$). It is further crucial that no reflections are cast onto the CubeStar lens as this might degrade the overall star detection performance.

CubeStar should be mounted so that it is pointing away from the Sun, Earth, and Moon during the nominal flight orientation. If these celestial bodies enter the FoV, images will over saturate thereby impeding star detection.

Although active thermal regulation is not required, it is recommended that CubeStar is mounted away from any solar rays. This will prevent performance degradation caused by excessive thermal conditions and possible damage to the image sensor.

CubeStar should be mounted so that there is enough slack on the wire harness. An excess in harness strain can lead to an unsecure connection or damage.



Care should be taken when mounting CubeStar. Ensure that the entire FoV is unobstructed and that there are no objects nearby that could reflect light into the lens. Any light falling on the lens except for the light of the stars will create an invalid image.

7.9.7 Axis Definition

Multi-axes sensors include the deployable magnetometer (deployed and undeployed state), redundant magnetometer, fine Sun sensor, fine nadir sensor and star tracker. The mounting of each of these sensors is specified by a set of three angles **alpha (α)**, **beta (β)**, and **gamma (γ)**. These three angles correspond to the angles in a Euler 3-2-1 rotation sequence to rotate the sensor's coordinate frame to the SBC. The sensor coordinate frames for CubeStar is defined as follows:

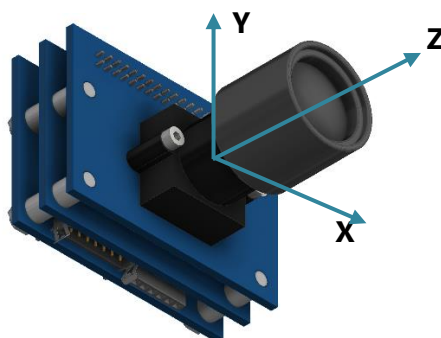


Figure 50 – CubeStar Sensor coordinate frame

The rotations α , β , and γ are around the sensor's Z, Y and X axes respectively.

7.9.8 Specifications

Specification	Value	Notes
Physical		
Mass	55.5g	Without mounting screws
Dimensions	50 x 35 x 55 mm	The length of about 55mm can change by ± 2 mm. (Without baffle)
	21 mm	Lens diameter
Electrical		
Operating Voltage	3.3V	
In-rush current peak	156 mA	Maximum inrush peak at power-on.
In-rush current time	1 ms	Maximum time of inrush peaks
Data Interface	I ² C and UART	
Average Power	142 mW	
Peak Power	264 mW	

Performance

Field of View	58° x 47°	Horizontal (X-axis) X Vertical (Y-axis) (Without baffle)
Star Catalogue Size	410	Subset of the Hipparcos Catalogue
Sensitivity Range	< 3.8	Star Magnitude
Sky cover:	99.71%	
Accuracy	<0.0154°(3σ)	Cross Axis (RA) [<0.0051°(1σ)]
	<0.0215°(3σ)	Cross Axis (DE) [<0.0072°(1σ)]
	<0.061°(3σ)	Roll [<0.0203°(1σ)]
Update Rate	1 Hz	
Max Tracking Rate	0.3°/s	
Max Acquisition Time	3 s	
Image Size	1.3 MB	1280x1030
Environmental		
Vibration	8.03 g	RMS random
Thermal	-10°C to +60°C	
Radiation	200MeV	High energy particle test

8 Document History

Version	Responsible person(s)	Pages	Date	Description of change
3.0	MK	ALL	20/03/2017	V3 First draft
3.01	MK	ALL	04/07/2017	Merge 3-Axis & Y-Momentum
3.02	MK	27,28	04/07/2017	Added CAN details
3.03	CJG	All	26/07/2017	General Updates
3.04	MK	All	27/07/2017	Accept/Reject team changes
3.05	MK	24 26	31/10/2017	Fixed incorrect I2C address Changed Table 8 to correct TLM (240)
3.06	CJG	32	02/02/2018	Removed Command 70
3.07	CJG	12	12/02/2018	Added description for the SPI bus
3.08	CCH	All	29/03/2018	General language editing
3.09	CCH	ALL	30/05/2018	Updated language editing
3.10	DGS	13	18/06/2018	Added warning about camera protrusion
3.11	DGS	13	29/10/2018	Specified pin length beneath stack and PC104 type at top
3.12	CJG	11	29/10/2018	Added Backup Power in Description
3.13	LV		20/01/2019	Added Relevant CubeACP version section Updated tables with Commands and Telemetries. See ACP Reference Manual for detail about interface changes.
3.14	CJG	39	24/01/2019	Extended the CubeStar accuracy description
3.15	CJG, HW, JG, DS, LV, GJR, WHS	All	27/09/2019	Large Update.
3.16	GJV, DGS	9, 22, 54	11/11/2019	Update PC104 table. Added uCN. Verified wheel balance spec
3.17	JG CJG	40 29,30	02/01/2020 16/03/2020	Update stowed magnetometer dimensions. Added explanation and figure for the method of epoxying the spacers.
3.18	CJG	24	26/03/2020	Updated Figure 6 with the correct position of BU_VIN